

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

This project explores the integration of **Intelligent Reflecting Surfaces (IRS)** with small-scale **MIMO wireless networks** to address the growing performance and energy-efficiency demands of next-generation communication systems. Targeting critical use cases such as IoT deployments, 5G/6G indoor hotspots, and urban coverage gaps, the IRS-assisted architecture leverages programmable metamaterials to dynamically manipulate electromagnetic wave propagation. Unlike traditional relay-based systems, IRS enables passive beamforming by adjusting the phase and amplitude of incident signals through tunable unit cells, without requiring active signal amplification.

When combined with compact 2×2 or 4×4 MIMO configurations, this approach delivers significant advantages: enhanced spectral efficiency via intelligent multipath control, mitigation of coverage dead zones in non-line-of-sight (NLOS) scenarios, and substantial reductions in power consumption. By transforming the wireless channel into a software-defined propagation environment, this technology represents a paradigm shift in how radio environments are engineered, offering scalable, low-power solutions for the high-density, high-throughput networks of the future.

Personalization: The system architecture includes adaptive control mechanisms that enable personalized wireless environments tailored to user-specific communication needs. Leveraging real-time beam steering and channel estimation powered by machine learning algorithms, the IRS-assisted MIMO system dynamically optimizes signal paths based on user location, device type, and traffic demand. For instance, IoT sensors requiring ultra-low power can be prioritized with energy-efficient paths, while high-throughput mobile users can benefit from optimized beamforming. This level of customization ensures that each user experiences consistent quality of service, making the system ideal for applications in smart homes, offices, and personalized IoT ecosystems.

Globalization: The project embraces globalization by designing a flexible, scalable, and region-agnostic communication architecture. The IRS-MIMO system supports multiple frequency bands, including sub-6GHz and mm Wave spectra, to meet regulatory and deployment requirements across different countries. Its compatibility with international wireless standards such as 5G NR and IEEE 802.11 ensures seamless integration into global infrastructure.

Furthermore, the use of software-defined radios and FPGA-based controllers enables rapid adaptation to regional needs, whether in densely populated urban centres, rural areas, or developing nations. This adaptability makes the technology a viable solution for global connectivity initiatives, including smart city rollouts and remote education.

Multimedia Services: IRS-assisted MIMO networks are especially well-suited for delivering high-quality multimedia services, including HD video streaming, real-time conferencing, AR/VR applications, and remote collaboration. By dynamically shaping the wireless channel and improving signal-to-noise ratio (SNR), the system reduces latency and jitter, critical factors for immersive media experiences.

The improved spectral efficiency and robust NLOS performance ensure consistent throughput in bandwidth-intensive applications, even in environments with multiple users and high interference levels. This makes the solution ideal for multimedia-rich settings like smart campuses, virtual event venues, and connected homes.

Software Implementation: The software implementation of this project is centered around COMSOL Multiphysics 6.3, which serves as the primary platform for simulating and analyzing the electromagnetic behavior of the IRS-MIMO system. COMSOL enables full-wave modeling of the metamaterial-based IRS structures, capturing complex physical effects such as near-field coupling, surface wave propagation, and frequency-dependent material properties.

The simulation environment is used to optimize the design of tunable unit cells, evaluate phase-shifting behavior, and analyze system-level performance under various propagation scenarios, including non-line-of-sight (NLOS) conditions. Coupled with

MATLAB and Python for control logic and post-processing, the simulation framework supports multi-physics co-simulation, integrating electromagnetic, thermal, and structural aspects to ensure robust, real-world performance. This comprehensive approach allows accurate prediction of system behavior and guides hardware design and prototyping decisions, ensuring that the software foundation directly translates into practical implementation success.

1.2 WIRELESS COMMUNICATION

Wireless communication is the transfer of information between two or more devices without the use of physical conductors, relying instead on electromagnetic waves to transmit signals across distances. It forms the backbone of modern connectivity, enabling technologies such as mobile networks, Wi-Fi, satellite systems, and IoT applications. In wireless systems, data is typically modulated onto carrier waves in various frequency bands (e.g., sub-6GHz, mm Wave) and transmitted through free space, where it may encounter obstacles, reflection, scattering, and interference.

The increasing demand for high-speed, low-latency, and energy-efficient communication in densely populated and dynamic environments has driven the evolution of wireless technologies. In the context of this project, wireless communication is enhanced through the integration of small-scale MIMO systems and Intelligent Reflecting Surfaces (IRS), which collaboratively manipulate signal propagation to improve coverage, spectral efficiency, and reliability without increasing transmission power or infrastructure complexity.

1.3 WIRELESS TECHNOLOGY

Wireless technology encompasses the various methods and standards used to transmit data over the air using electromagnetic signals, eliminating the need for physical connections like cables or fibre optics. It includes a wide range of applications—from mobile networks (2G to 6G), Wi-Fi (IEEE 802.11), and Bluetooth, to satellite communications and emerging IoT protocols such as LoRa and Zigbee.

The evolution of wireless technology has been marked by advancements in bandwidth efficiency, data rates, latency reduction, and energy optimization. Modern wireless technologies rely heavily on techniques such as Multiple Input Multiple Output (MIMO), Orthogonal Frequency Division Multiplexing (OFDM), and beamforming to enhance performance in complex environments. In this project, wireless technology is further advanced through the implementation of Intelligent Reflecting Surfaces (IRS), which act as programmable passive elements that intelligently control signal reflection, enabling dynamic reconfiguration of the wireless environment. This synergy with small-scale MIMO systems enables a new generation of adaptive, efficient, and high-performance wireless communication networks suitable for next-generation applications like 5G/6G, smart cities, and industrial automation.

1.4 WIRELESS COMMUNICATION STANDARDS

Wireless communication standards are a set of protocols and guidelines that govern the design, operation, and interoperability of wireless communication systems. These standards ensure that devices, networks, and systems can communicate effectively and reliably across different technologies and platforms. The most widely adopted standards in modern wireless communication include 2G, 3G, 4G LTE, and 5G for mobile networks, Wi-Fi (IEEE 802.11) for local area networks, and Bluetooth (IEEE 802.15.1) for short-range communication. Each standard defines key aspects such as frequency bands, modulation techniques, data rates, and error correction methods to optimize performance.

The evolution of these standards has been driven by the increasing demand for higher data rates, lower latency, and more reliable connectivity, particularly in dense environments and emerging applications such as IoT and smart cities. The transition to 5G and the future of 6G involve new technologies like Massive MIMO, beamforming, and Intelligent Reflecting Surfaces (IRS), which aim to address the limitations of current systems. In this context, the integration of IRS with MIMO systems represents a key innovation to enhance spectral efficiency, network capacity,

and user experience, aligning with the next generation of wireless communication standards.

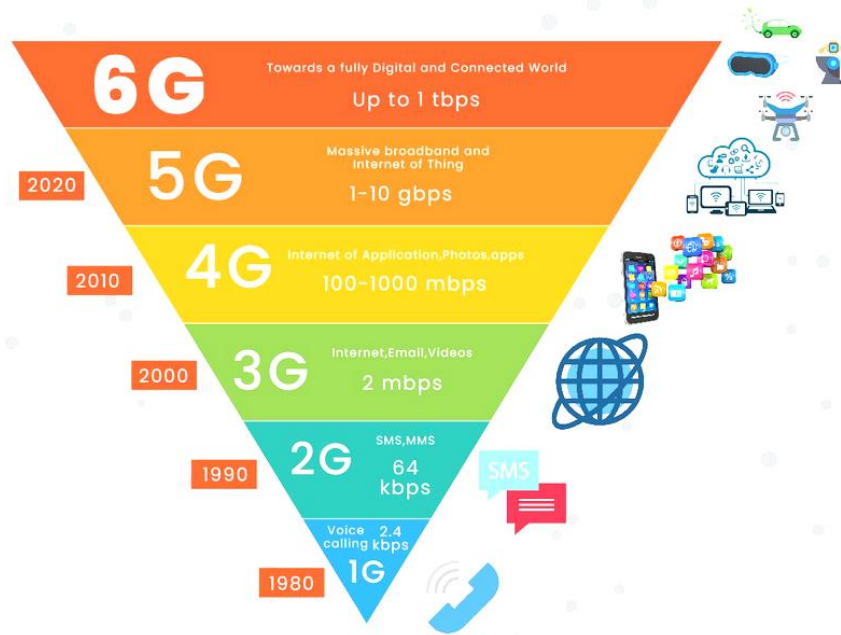


Fig 1.4 EVALUATION OF 1G TO 6G WIRELESS COMMUNICATION

1.5 5G/6G GENERATION

The 5G generation represents a major leap forward in wireless communication, designed to meet the demands of high-speed internet, low latency, and massive connectivity required by applications such as autonomous vehicles, smart cities, and the Internet of Things (IoT). It introduces several key innovations, including massive MIMO (Multiple Input Multiple Output), beamforming, network slicing, and millimetre-wave (mm Wave) technology. These advancements allow 5G networks to support ultra-high-speed data rates (up to 10 Gbps), significantly reduced latency (as low as 1 ms), and the capacity to connect millions of devices in a given area, facilitating the growth of IoT ecosystems and connected environments. Furthermore, 5G enables low latency communication crucial for mission-critical applications like industrial automation and remote surgery.

Looking ahead, 6G will usher in an even more transformative era of wireless communication. It aims to build upon 5G's capabilities and will likely support data rates in excess of 100 Gbps, ultra-reliable low-latency communication (URLLC), and

massive machine-type communications (mMTC). 6G is expected to integrate terahertz (THz) communication, AI-driven network management, and holographic communications for applications such as immersive augmented and virtual reality (AR/VR), smart healthcare, and real-time 3D mapping. Additionally, Intelligent Reflecting Surfaces (IRS) and reconfigurable intelligent surfaces (RIS) will play a crucial role in the 6G landscape by enabling flexible and adaptive wireless environments that enhance signal quality and reduce interference, leading to better user experience and network efficiency. The transition from 5G to 6G will focus on increasing both the capacity and intelligence of the network to accommodate the ever-growing demand for ubiquitous, high-speed, low-latency connectivity across diverse environments.

1.6 MIMO AND TYPES

Multiple Input Multiple Output (MIMO) is a crucial technology in modern wireless communication that utilizes multiple antennas at both the transmitter and receiver to improve data throughput, signal reliability, and network capacity. MIMO works by exploiting multipath propagation, where signals bounce off surfaces, allowing multiple versions of the same signal to reach the receiver. The receiver then separates these signals to increase data transmission without needing additional spectrum. MIMO is particularly useful in high-demand environments, such as urban and indoor spaces, where signal degradation and interference are common. It is a cornerstone technology for 4G LTE, 5G, and beyond.

There are several types of MIMO systems, each suited to different communication environments and performance requirements:

Single-User MIMO (SU-MIMO): This type focuses on providing high throughput for a single user by utilizing multiple transmit and receive antennas to send and receive multiple data streams simultaneously. It improves the data rate and reliability for users in areas with poor signal conditions, such as urban and indoor environments.

Multi-User MIMO (MU-MIMO): Unlike SU-MIMO, which serves one user at a time, MU-MIMO allows multiple users to share the same time-frequency resources. It increases network capacity and overall efficiency by serving several users simultaneously in a cellular environment, reducing congestion and improving system performance.

Massive MIMO: This is an advanced version of MIMO that employs a large number of antennas (often hundreds or thousands) at the base station. Massive MIMO boosts spectral efficiency, reduces interference, and improves coverage, making it especially useful for 5G and future 6G networks. It can support hundreds of users concurrently with improved signal quality and reliability.

Spatial MIMO: This type of MIMO uses the spatial diversity provided by multiple antennas to send signals through different spatial channels. It is beneficial for improving the signal-to-noise ratio (SNR) in environments with high interference or fading.

Each of these MIMO types contributes to the advancement of wireless communication systems, particularly as the demand for high-speed, reliable, and scalable networks continues to grow. The integration of MIMO with technologies such as Intelligent Reflecting Surfaces (IRS) further enhances performance by improving signal propagation, beamforming, and coverage, especially in complex and dynamic environments.

1.7 IRS DETAILS

Intelligent Reflecting Surfaces (IRS) are an emerging technology in wireless communication that offer a novel way to enhance signal propagation by intelligently controlling the reflection of electromagnetic waves in the environment. IRS consists of large arrays of passive, reconfigurable elements that can adjust the phase and amplitude of incoming signals without the need for active amplification or transmission power. These surfaces are typically composed of meta-materials, which are engineered to manipulate electromagnetic waves in a controlled manner, allowing for dynamic beam steering and signal optimization. IRS devices can be made of cost-

effective components such as PIN diodes or varactors, which enable precise control of signal reflection and phase shifting.

The key advantage of IRS lies in its ability to transform the wireless environment into a software-defined space. By reconfiguring the propagation path of signals, IRS enhances spectral efficiency, signal-to-noise ratio (SNR), and coverage while reducing interference. This makes it particularly valuable for non-line-of-sight (NLOS) conditions, where traditional communication systems may suffer from poor signal quality due to obstacles like buildings or walls. IRS also offers significant improvements in energy efficiency compared to traditional relay systems, as it consumes much less power by passive reflection rather than active signal amplification.

In practice, IRS can be deployed in various environments such as urban settings, indoor areas, and smart factories, where it helps mitigate common challenges like multipath fading and signal blockages. When integrated with small-scale MIMO systems, IRS significantly boosts the area spectral efficiency, improving throughput and capacity for IoT, 5G, and even emerging 6G networks. The integration of IRS with beamforming and machine learning algorithms enables real-time optimization of wireless channels, adapting to dynamic conditions and user mobility. In the long term, IRS is expected to play a pivotal role in the development of future wireless communication systems by enabling flexible, high-capacity, and energy-efficient networks.

1.8 COMSOL MULTIPHYSICS 6.3

COMSOL Multiphysics 6.3 is a powerful simulation software platform used for modelling and simulating a wide range of physical phenomena, including electromagnetics, fluid dynamics, structural mechanics, and heat transfer. It is particularly well-suited for multi-physics simulations, where interactions between different physical fields are critical. In the context of this project, COMSOL Multiphysics is utilized to model and analyse the performance of Intelligent

Reflecting Surfaces (IRS) integrated with small-scale MIMO systems for wireless communication applications.

Using COMSOL, we can conduct full-wave electromagnetic simulations to evaluate the behaviour of IRS structures, which are composed of reconfigurable meta-material elements that manipulate electromagnetic waves. The software allows us to study the effects of phase shifting, signal reflection, and beamforming at various frequencies, such as sub-6 GHz and millimetre-wave (mm Wave) bands, which are key for IoT and 5G/6G networks. COMSOL's RF Module is particularly useful for simulating the propagation of electromagnetic waves in complex environments, taking into account near-field interactions, material dispersion, and the impact of obstacles or non-line-of-sight conditions.

The parametric sweep and optimization tools in COMSOL Multiphysics 6.3 enable the fine-tuning of IRS configurations to maximize performance metrics such as spectral efficiency, signal-to-noise ratio (SNR), and coverage. Additionally, electromagnetic wave coupling can be simulated in the presence of real-world constraints like antenna placement, environmental factors, and channel characteristics. The software's user-friendly interface, combined with its powerful solvers, allows for iterative testing and modification of design parameters, ensuring a comprehensive and efficient approach to the development of IRS-assisted MIMO systems.

By using COMSOL Multiphysics 6.3 for system-level simulations, we are able to validate the theoretical predictions, assess potential design issues, and optimize IRS configurations for practical deployment in real-world environments, such as indoor and urban settings where wireless performance is often hindered by interference and signal degradation.

1.9 SCOPE OF THE PROJECT

The scope of this project centres on the design, simulation, and evaluation of an **Intelligent Reflecting Surface (IRS)-assisted small-scale MIMO wireless communication system** using **COMSOL Multiphysics 6.3**. The primary objective is

to explore how IRS technology can significantly enhance wireless signal propagation, spectral efficiency, and energy performance in environments where traditional MIMO systems face challenges, such as indoor spaces, urban areas, and IoT-dense networks. The project covers the modelling of tunable IRS elements, full-wave electromagnetic simulation of wave interactions, and performance analysis under different propagation conditions.

It also investigates the integration of IRS with 2×2 and 4×4 MIMO configurations, focusing on metrics such as signal-to-noise ratio (SNR), coverage improvement, and system power consumption. The scope extends to practical considerations like hardware feasibility, phase shift quantization, and real-time channel adaptation using intelligent control mechanisms.

Overall, this project contributes to the advancement of 5G and future 6G technologies by demonstrating the potential of IRS to transform passive wireless environments into dynamically controllable communication channels.

1.10 OBJECTIVE

The primary objective of this project is to design and implement an **Intelligent Reflecting Surface (IRS)-assisted small-scale MIMO wireless communication system** to improve wireless signal performance in complex environments.

The project aims to demonstrate how IRS can enhance **spectral efficiency**, **signal coverage**, and **energy efficiency** by intelligently manipulating the propagation of electromagnetic waves.

Using **COMSOL Multiphysics 6.3**, the project seeks to model IRS structures, analyse wave reflection and phase control, and optimize the overall system performance under various channel conditions. Additional objectives include reducing power consumption, addressing non-line-of-sight (NLOS) communication challenges, and ensuring compatibility with existing 5G/6G and IoT infrastructures.

CHAPTER 2

LITERATURE SURVEY

This chapter gives technical survey on some previous works. Some concepts related to the proposed work are mentioned as follows.

2.1 Qingqing Wu, Shuowen Zhang, Beixiong Zheng, Changsheng You, and Rui Zhang, “**Intelligent Reflecting Surface-Aided Wireless Communications**” (2021).

Objective:

The paper aims to explore **Intelligent Reflecting Surfaces (IRS)** as a transformative technology for improving wireless communication in future networks like **5G and 6G**. It focuses on optimizing passive beamforming, improving channel estimation, and enabling energy-efficient deployment.

Methodology:

A tutorial-based approach is used, covering theoretical signal models, IRS hardware behaviour, and system integration. The authors apply optimization techniques (e.g., alternating optimization) to configure IRS phase shifts and transmit beamforming, with analysis across SISO, MIMO, and OFDM systems.

Performance Measured:

Key performance metrics include **SNR, achievable rate, power efficiency, and beamforming gain**. The study shows that IRS can significantly enhance signal strength and reduce energy consumption, especially in non-line-of-sight (NLoS) conditions.

Suggestions:

Future research should address more realistic channel models, precise IRS control hardware, and low-complexity algorithms for dynamic beamforming. Integration with **THz communication, AI-based control, and massive MIMO** is recommended for future 6G systems.

2.2 Wei Lu, Bin Deng, Qiqing Fang, Xiaoqiao Wen, and Shixin Peng, “**Intelligent Reflecting Surface-Enhanced Target Detection in MIMO Radar**” (2021).

Objective:

This paper investigates the use of **Intelligent Reflecting Surfaces (IRS)** to enhance target detection in **collocated MIMO radar systems**. The main objective is to improve received signal power and detection accuracy by optimizing IRS-assisted beamforming, thus enabling sharper angle estimation and better spatial resolution in radar applications.

Methodology:

The authors propose an **IRS-assisted target detection algorithm** that optimizes IRS phase shifts based on SNR criteria using a practical hardware model. They use a modified **Amplitude and Phase Estimation (APES)** method for target detection and derive closed-form solutions for suboptimal IRS configurations. The study includes detailed channel modelling with Rician fading and a CRB-based performance analysis.

Performance Measured:

Performance is evaluated using **Cramér–Rao Bound (CRB)**, **SNR**, **detection probability**, and **spatial spectral resolution**. Simulations show that the IRS-enhanced radar achieves better detection accuracy and finer angle resolution, especially as the number of IRS elements increases. A lower Rician factor (closer to Rayleigh fading) yields better performance due to reduced channel degradation.

Suggestions:

The study suggests using **Rayleigh channels** for IRS-to-receiver links to maximize performance. It also recommends minimizing control overhead via **discrete phase shifts** and emphasizes the importance of practical IRS hardware modelling. Future work may explore IRS placement strategies and broader applications in adaptive radar systems.

2.3 Zheng Li, Zhengyu Zhu, Zheng Chu, Yingying Guan, De Mi, Fan Liu, Lie Liang Yang, “**IRS-Assisted Integrated Sensing and Wireless Power Transfer**” (2023).

Objective:

This paper investigates an **IRS-assisted integrated sensing and wireless power transfer (ISWPT)** system aimed at smart transportation infrastructure. The objective is to simultaneously achieve efficient **target sensing** and **RF energy harvesting** by optimizing the system’s beamforming and IRS phase shifts under practical constraints.

Methodology:

The authors formulate a non-convex optimization problem to balance the trade-off between sensing accuracy and harvested energy, using a **trade-off factor (ρ)**. Two solutions are proposed: a **semi-definite programming (SDP)** method and a **low-complexity (LC)** algorithm that combines **successive convex approximation (SCA)** for beamforming and **majorization-minimization (MM)** for phase control. These solutions address the coupling between transmit and reflection variables.

Performance Measured:

Key performance metrics include **sum harvested energy**, **beampattern gain**, and **algorithm convergence rate**. Simulations show that both SDP and LC methods converge rapidly (within 2–3 iterations) and significantly outperform baseline methods without IRS or with random phase shifts. The harvested energy increases with IRS element count, and optimal phase tuning yields sharp target localization.

Suggestions:

The authors recommend deploying more IRS elements for better resolution and energy performance. They also highlight the importance of phase shift optimization and suggest future studies on **adaptive IRS placement**, **dynamic environmental modelling**, and **joint waveform design** for enhanced ISWPT performance.

2.4 Yuan Zheng, Suzhi Bi, Ying Jun (Angela) Zhang, Zhi Quan, and Hui Wang, “**Intelligent Reflecting Surface Enhanced User Cooperation in Wireless Powered Communication Networks**” (2020).

Objective:

The paper investigates the use of **Intelligent Reflecting Surfaces (IRS)** in **Wireless Powered Communication Networks (WPCNs)** to improve throughput. It focuses on optimizing the IRS phase shifts, power, and time allocation to enhance energy harvesting and data transmission between a **High-Altitude Platform (HAP)** and **Wireless Devices (WDs)**.

Methodology:

The system model involves two WDs cooperating via IRS to relay signals. The problem is formulated as an optimization task to maximize throughput, which is solved using **Semidefinite Programming (SDP)** and convex optimization tools like **CVX**. Rank-one constraints are relaxed, and suboptimal solutions are derived to recover phase shifts and power allocations.

Performance Measured:

Simulation results show that IRS-assisted cooperation provides up to **652%** higher throughput than conventional methods. The performance improves with more IRS elements, but the gains decrease when inter-user channels weaken. Different distances between HAP and WDs are evaluated, showing the effectiveness of IRS in mitigating weak channel conditions.

Suggestions:

Future work can focus on **multi-user WPCNs**, where users share energy and relay information. Challenges like **discrete-phase IRS** and real-time optimization need further exploration for practical implementations. The integration of IRS in **5G/6G** could enhance network efficiency and reduce costs.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1 EXISTING SYSTEM

In traditional Wireless Sensor Networks (WSNs), data transmission systems often suffer from limitations that affect performance and reliability. These networks, made up of multiple sensor nodes, are used to monitor environmental parameters but face challenges like interference, fading, and signal blockage—especially in urban or remote areas. Multipath fading caused by obstacles can distort signals and cause delays.

Conventional WSNs also have restricted data rates and low spectral efficiency due to limited bandwidth and basic communication protocols. Most sensor nodes operate on low power, which limits their ability to handle high data traffic or real-time applications. Battery limitations and energy waste through retransmissions or idle listening further reduce network lifespan.

Moreover, WSNs lack the adaptability to changing conditions, as they often cannot reconfigure intelligently or optimize transmission. These drawbacks impact accuracy, increase latency, and reduce scalability in larger or mission-critical deployments.

3.1.1 DISADVANTAGE

- **High Energy Consumption**

Base stations and towers consume significant power, especially in rural or remote areas.

- **Coverage Gaps**

Difficult to provide reliable connectivity in complex environments like tunnels, basements, or mountainous regions.

- **High Deployment Cost**

Installing and maintaining base stations and towers is expensive.

- **Signal Blockage**

Obstacles such as buildings and terrain reduce signal quality through reflection and absorption.

- **Interference Issues**

Crowded frequency bands cause signal degradation due to overlapping transmissions.

- **Inefficient Spectrum Use**

Traditional networks cannot dynamically allocate spectrum based on real-time needs.

- **Latency Problems**

Long signal paths and congestion lead to delays in data transmission.

- **Environmental Impact**

Tower construction and electronic waste pose environmental challenges.

- **Security Vulnerabilities**

Centralized infrastructure increases exposure to cyberattacks and data breaches.

- **Scalability Limitations**

Hard to scale in growing urban or industrial environments without large infrastructure upgrades.

- **Limited Adaptability**

Traditional WSNs struggle to adapt to dynamic environmental or network conditions, resulting in poor performance under variable loads or in changing terrains.

- **Maintenance Complexity**

Regular maintenance and troubleshooting of sensor nodes and infrastructure can be labour-intensive and costly, especially in inaccessible or hazardous areas.

3.2 PROPOSED SYSTEM

The **proposed system** integrates **Multiple-Input Multiple-Output (MIMO)** antenna-equipped sensor nodes within a **wireless network**, **strategically distributed** to **monitor** and **optimize signal transmission** in **challenging environments**. These nodes are capable of **measuring signal strength**, **Channel State Information (CSI)**, and **interference patterns** in **real time**. Additionally, **RF sensors** are deployed to collect **detailed data** on **multipath propagation**, **fading conditions**, and other **environmental effects** that typically degrade **signal quality**.

To counteract these impairments, the system leverages **advanced signal processing techniques** such as **beamforming** and **precoding**, which allow for **directed signal transmission**, **reducing interference** and **improving reliability**. **Machine learning models** are incorporated to **detect**, **classify**, and **mitigate interference dynamically**. Furthermore, the system enables the **real-time adjustment of antenna parameters**, allowing it to **respond adaptively** to **environmental changes** and **optimize communication paths** for improved **network efficiency**.

This **intelligent, adaptive architecture** represents a **significant step** toward **next-generation wireless networks**, offering enhanced **performance**, **reliability**, and **energy efficiency**.

3.2.1 ADVANTAGE

- **Enhanced Signal Quality**

Real-time monitoring of signal strength and CSI allows for optimized data transmission, minimizing packet loss and distortion.

- **Improved Spectral Efficiency**

Beamforming and precoding techniques focus energy where it's needed, reducing spectrum wastage and enhancing throughput.

- **Adaptive Interference Mitigation**

Machine learning-based detection enables proactive management of interference sources, which is difficult in static traditional systems.

- **Better Multipath Handling**

MIMO and RF sensing allow the system to exploit or mitigate multipath effects rather than suffer from them, leading to higher reliability.

- **Dynamic Network Reconfiguration**

The system can adapt antenna orientation, power, and signal parameters in real time, something fixed tower-based systems cannot do.

- **Energy Efficiency**

Focused beam transmission and fewer retransmissions help reduce the power consumption of sensor nodes, extending operational life.

- **Scalability**

The modular design and intelligent coordination allow easy network expansion without sacrificing performance.

- **Reduced Latency**

By optimizing transmission paths and avoiding congestion zones through learning-based routing, data reaches the destination faster.

- **Smaller Physical Footprint**

Unlike towers, this system can be deployed on existing infrastructure (walls, poles, buildings), minimizing visual and environmental impact.

- **Resilience to Environmental Changes**

The system continuously learns and adapts to new channel conditions, making it robust to terrain or weather-related disruptions.

- **Improved Security**

The use of adaptive transmission strategies and decentralized control points helps reduce vulnerability to interception and jamming, enhancing overall network security.

- **Intelligent Resource Allocation**

Machine learning algorithms optimize resource usage—such as bandwidth, power, and channel assignment—based on real-time demand and network conditions.

CHAPTER 4

SYSTEM DESIGN

4.1 BLOCK DIAGRAM

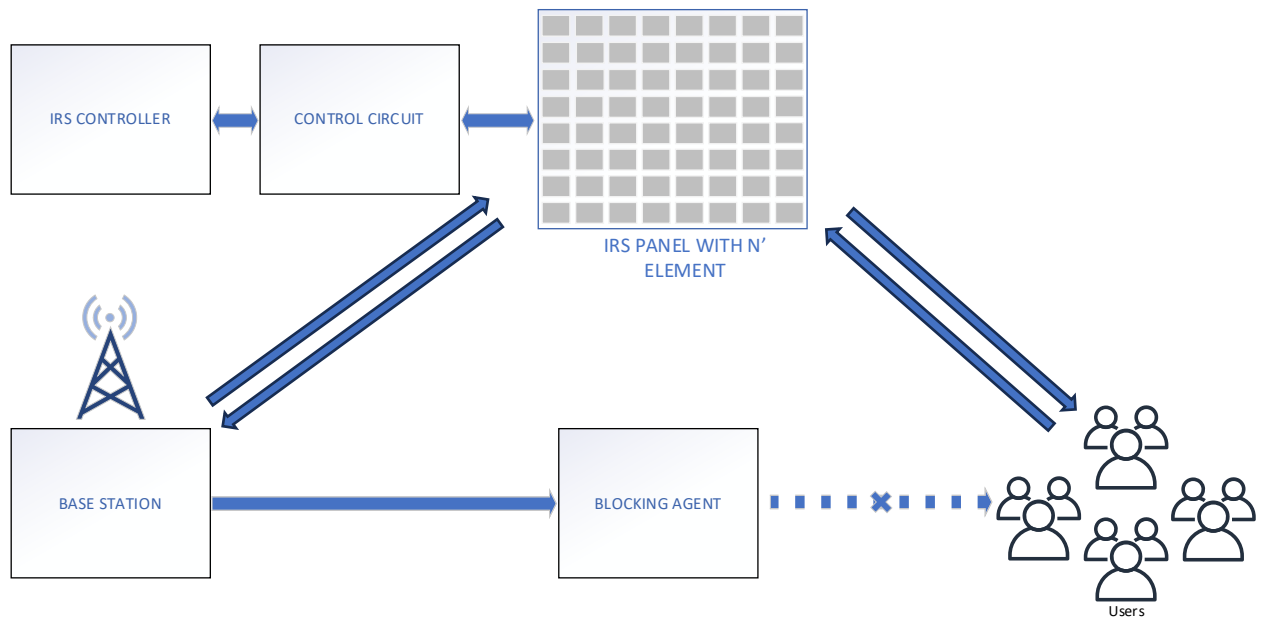


Fig 4.1 BLOCK DIAGRAM

4.2 DESCRIPTION

1. IRS Controller and Control Circuit

The operation of the intelligent reflecting surface (IRS) system starts with the IRS Controller, which is typically implemented using a microcontroller or embedded processor such as a Raspberry Pi or FPGA. This component is tasked with monitoring, processing, and issuing configuration commands to control the reflective properties of the IRS panel. It serves as the brain of the IRS system, enabling adaptive beamforming and signal steering based on real-time channel conditions and user locations.

Connected directly to the IRS Controller is the Control Circuit, which converts the digital control signals into the required analog signals (such as voltage levels) that adjust the phase or amplitude response of each reflecting element in the IRS panel. This circuit ensures that each of the N reflecting elements on the IRS panel can be finely tuned for optimal performance. The dynamic control mechanism allows the IRS to function intelligently and flexibly in a changing wireless environment, such as user mobility, signal blockage, or interference.

2. IRS Panel with N Elements

The IRS Panel, consisting of N discrete reflecting elements, acts as a passive but intelligent medium to manipulate incoming wireless signals. These elements are designed with tunable components (e.g., PIN diodes, varactors, or MEMS switches) that allow control over the phase, amplitude, or polarization of reflected signals. The elements are arranged in a grid-like fashion to form a reconfigurable meta surface.

When signals from the base station impinge upon the IRS panel, the Control Circuit—under the supervision of the IRS Controller—modifies the surface reflection properties of each IRS element to form a directional beam toward intended user locations. This process is often referred to as passive beamforming. Unlike traditional active antennas, the IRS does not require power-hungry RF chains or signal generation, making it a highly energy-efficient solution for enhancing signal coverage and reliability.

Thus, the IRS Panel serves a dual purpose: (1) it receives control instructions from the IRS Controller to reconfigure its elements, and (2) it reflects the incident signals from the base station in desired directions to serve users, even in non-line-of-sight (NLOS) conditions.

3. Base Station

The Base Station (BS) is the central transmission point in the wireless communication system. It generates and transmits downlink signals intended for multiple users. However, in dense urban or indoor environments, direct communication paths between the base station and users may be partially or fully obstructed, limiting the base station's ability to maintain a reliable link with all users.

To overcome this, the base station also transmits signals towards the IRS panel, which acts as a signal repeater or reflector. By carefully designing the signal path through the IRS, the system can circumvent physical obstacles and provide indirect but optimized communication links to users. The base station and IRS thus work in tandem—one actively transmitting signals and the other intelligently redirecting them.

4. Blocking Agent

Between the base station and the users exists a Blocking Agent, such as a wall, building, tree, or any large object that obstructs the Line-of-Sight (LOS) signal path. This blocking agent introduces a Non-Line-of-Sight (NLOS) communication environment, which degrades signal strength and quality. Users located behind the blockage may experience low signal-to-noise ratio (SNR), increased latency, and lower throughput, ultimately impairing the quality of service.

This challenge is what motivates the use of IRS technology. While traditional methods like signal relays or higher transmission power may provide limited improvements, they also consume more energy and resources. The IRS, by contrast, offers a passive and energy-efficient alternative to re-establish communication links disrupted by the blocking object.

5. Users

The Users, represented as end devices such as smartphones, IoT nodes, or laptops, are the final recipients of the transmitted signals. In scenarios where direct signal paths are blocked, the users rely on the IRS to receive a strong and coherent signal. The IRS reflects and reshapes the signal beams such that the signal energy is constructively directed toward the user locations, compensating for path loss and improving signal coverage.

This re-routed signal path ensures that users who are in a shadow region (due to the blocking object) can still achieve high data rates and reliable connectivity, without requiring additional infrastructure. The ability of the IRS to adapt its reflection pattern in real time allows it to support multiple users simultaneously, each potentially requiring a unique signal path.

System Summary

The entire system operates in a synchronized manner to mitigate the effects of physical obstructions and enhance wireless communication. The Base Station transmits the original signal, which is intercepted by a Blocking Agent and fails to reach users directly. Simultaneously, this signal is also directed towards the IRS Panel, which, under the guidance of the IRS Controller and Control Circuit, reconfigures its elements to reflect the signal toward the intended Users. As a result, the system enables seamless and energy-efficient communication in otherwise unreachable NLOS scenarios.

CHAPTER 5

HARDWARE REQUIREMENTS

The hardware requirements and their data sheets needed in our projects is given below

- Capacitor 3pF
- Inductor 6nH
- Varactor Diode
- IRS Panel with Tuneable Elements
- RF Transmitter & Receiver
- Wi-Fi Transmitter & Receiver
- Tower
- Antenna
- Battery
- FPGA

These are the hardware tools required to get the result of reducing multiple antennas. Their descriptions are follows:

5.1 CAPACITOR (3 pF)

A capacitor stores electrical energy and is widely used for filtering, tuning, and coupling signals. A 3-pF capacitor is very small and mainly used in high-frequency circuits like RF and microwave systems to precisely control signal behaviour.



Fig 5.1 Capacitor

In IRS unit cells, 3 pF capacitors help tune the resonant frequency, enabling precise phase control of reflected waves. This tuning improves beam steering and signal efficiency by adjusting how the surface reflects wireless signals.

5.2 INDUCTOR (6 nH)

An inductor stores energy in a magnetic field and is used in tuning and filtering circuits. A 6 nH inductor is very small and used at high frequencies to shape signal response and create resonant circuits with capacitors.



Fig 5.2 Inductor

The 6 nH inductor works with capacitors in each IRS unit cell to control the phase and amplitude of reflected signals. Together, they form LC circuits that enable dynamic beam steering and help filter unwanted frequencies for clearer signals.

5.3 VARACTOR DIODE

A varactor diode acts as a voltage-controlled capacitor, with capacitance changing based on applied reverse voltage. It is used in tuning circuits like oscillators and filters without mechanical parts.



Fig 5.3 Varactor Diode

Varactor diodes allow real-time tuning of IRS unit cells by varying capacitance through control voltage, changing the phase of reflected waves. This makes the IRS adaptable, enabling dynamic beamforming and improving wireless communication without moving parts.

5.4 IRS PANEL

An **IRS (Intelligent Reflecting Surface) panel with tunable elements** is a key component in next-generation wireless communication systems, particularly in 6G, mm Wave, and IoT networks. It is designed to control the propagation of electromagnetic waves in the environment **without requiring active transmission**. Instead, it **manipulates the reflection** of incident signals using passive or semi-passive electronic elements integrated into its surface.

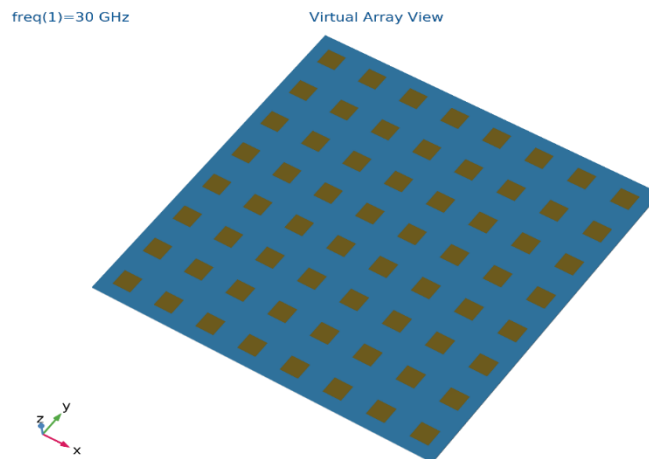


Fig 5.4 IRS Panel

What is an IRS Panel?

An IRS panel is a **planar array** made up of many tiny units called **meta-atoms** or **unit cells**, typically organized in a 2D grid. Each unit cell acts like a small antenna or reflector that can **adjust the phase, amplitude, or polarization** of the reflected electromagnetic wave.

Tunable Elements in IRS

The **tunable elements** embedded within each unit cell allow the IRS to dynamically alter the way it reflects signals. These components include:

- **Varactor diodes** (voltage-controlled capacitors)
- **PIN diodes** (used for switching on/off or between states)
- **MEMS switches** (micro-electromechanical systems)
- **Graphene-based or ferroelectric materials** (for phase change properties)

Each element responds to a **control signal** (usually a DC bias voltage) from an external controller, changing its electrical properties and, thus, the behaviour of the reflected signal.

Applications in Wireless Networks

- **Coverage Extension:** Redirect signals into dead zones or hard-to-reach areas (e.g., around buildings or obstacles).
- **Interference Management:** Cancel or redirect interfering signals to improve signal-to-noise ratio.
- **Energy Efficiency:** Unlike traditional relays or repeaters, IRS panels are mostly passive and consume very little power.
- **Security:** IRS can reflect signals selectively to limit eavesdropping risks.
- **Beamforming:** Works alongside MIMO systems to enhance directional transmission.

Key Advantages

- **Low Power Consumption:** Passive or semi-passive components require minimal energy.
- **Cost-Effective Deployment:** Can be installed on walls, ceilings, or infrastructure without active RF components.

- **Real-Time Reconfigurability:** Enables adaptive behaviour based on changing network conditions.
- **Compact Design:** Easily integrated into indoor or outdoor environments.

Example: Using Varactor Diodes

If each unit cell contains a **varactor diode**, applying a different bias voltage changes its capacitance. This, in turn, shifts the resonant frequency and **alters the phase** of the reflected wave. By tuning these diodes across the panel, the entire surface can control the reflection pattern of RF waves dynamically.

5.5 RF TRANSMITTER & RECEIVER

An **RF (Radio Frequency) Transmitter and Receiver** system is used to **send and receive data wirelessly** through radio waves. It is a fundamental part of wireless communication systems such as Wi-Fi, Bluetooth, GPS, RFID, remote controls, and IoT devices.



Fig 5.5 RF Transmitter and Receiver

5.5.1 RF TRANSMITTER

An **RF (Radio Frequency) transmitter** is a crucial component in wireless communication systems. It is responsible for converting data—such as audio, video, or digital signals—into electromagnetic waves that can travel through the air to a remote receiver. This enables devices to communicate wirelessly without the need for physical connections.

The process starts with an **oscillator**, which generates a stable carrier signal at a fixed radio frequency. This carrier wave acts as the foundation for transmitting data. A **modulator** then impresses the actual information onto the carrier wave by altering its amplitude, frequency, or phase, depending on the chosen modulation technique. Common modulation methods include AM, FM, PSK, and FSK.

Once the data is modulated, it is sent through a **power amplifier** to increase the strength of the signal. This is necessary to ensure the signal can travel long distances without significant loss or distortion. The amplified signal is then fed to an **antenna**, which converts the electrical signal into electromagnetic waves and radiates them into the air.

RF transmitters are used in a wide range of applications including mobile phones, Wi-Fi networks, radio broadcasting, satellite communication, and IoT systems. They provide the essential link that enables wireless data transfer between devices in personal, commercial, and industrial environments.

5.5.2 RF RECEIVER

An **RF receiver** is the counterpart to the RF transmitter. It captures radio signals from the environment and processes them to recover the original information that was transmitted. Receivers play a vital role in any wireless system, as they make it possible to decode and use the data sent through radio waves.

The first stage of reception involves an **antenna**, which picks up the electromagnetic wave and converts it back into a weak electrical signal. This signal often contains unwanted noise and interference, so it is passed through a **band-pass filter** that isolates the desired frequency range while rejecting others.

After filtering, the signal enters a **low-noise amplifier (LNA)**. This component boosts the strength of the weak incoming signal with minimal addition of noise, improving the clarity and quality of the signal for further processing. Next, a **demodulator** extracts the original data from the modulated carrier wave. Depending on how the data was modulated by the transmitter, different demodulation

techniques are used. Finally, a **decoder** converts the demodulated signal into a usable format, such as audio, video, or digital data.

RF receivers are widely used in everyday devices like smartphones, televisions, radios, and wireless sensors. They are also essential in advanced communication systems, where they must function reliably in environments with interference, multiple signals, and variable conditions. In modern designs, receivers can be enhanced using technologies like **Intelligent Reflecting Surfaces (IRS)**, which help redirect and strengthen signals for better performance in complex wireless environments.

5.6 TOWER

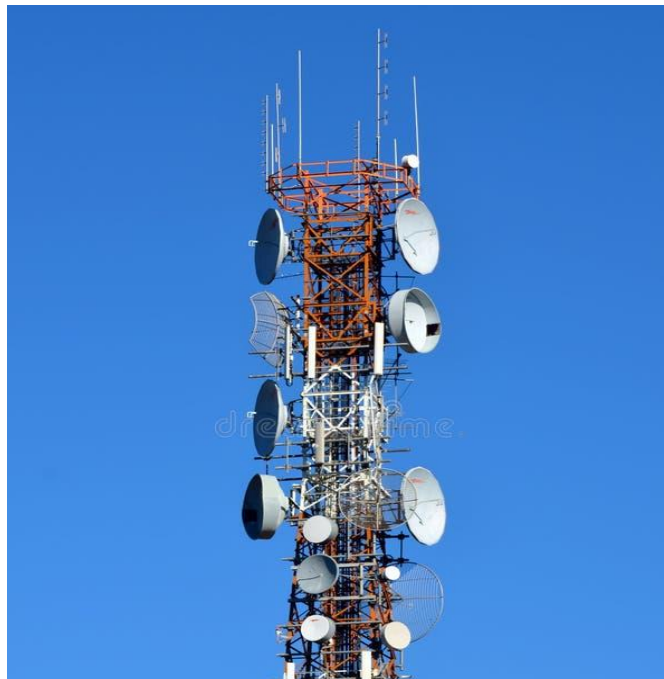


Fig 5.6 Communication Tower

A communication tower is an essential infrastructure component in wireless communication systems, designed primarily to elevate antennas and other transmission devices above the surrounding terrain to maximize signal coverage and quality. These towers enable effective transmission of radio frequency (RF) signals for cellular networks, television broadcasting, radio stations, and emergency communication services.

The structural design of communication towers must consider several critical factors, including material selection, load-bearing capacity, wind resistance, and seismic stability. Most towers are constructed using galvanized steel due to its high strength-to-weight ratio and resistance to corrosion, which ensures longevity and minimal maintenance.

The height of the tower is a significant parameter that influences the line-of-sight range and overall network efficiency, necessitating careful site surveys and engineering calculations before installation. Furthermore, safety features such as lightning protection, grounding systems, and access ladders are integrated into the design to safeguard both the equipment and maintenance personnel.

This project focuses on the comprehensive design and analysis of a communication tower to optimize its structural integrity and functional performance, supporting the deployment of advanced wireless technologies such as 5G and Internet of Things (IoT) networks. Through finite element modelling and simulation, the tower's response to dynamic loads such as wind gusts and seismic forces is evaluated to ensure compliance with international engineering standards.

5.7 ANTENNA



Fig 5.7 Antenna

An antenna is a fundamental element in any wireless communication system, serving as the interface between the radio frequency (RF) circuitry and the free space through which signals propagate. Its primary function is to efficiently radiate RF energy into the environment during transmission and to collect RF energy during reception with minimal loss. The design of an antenna depends heavily on its intended frequency band, application, and operational environment.

Antennas are classified into various types based on their shape, size, and radiation characteristics. Common types include dipole antennas, which are simple and widely used for many applications; monopole antennas, which are a variation of dipoles mounted above a ground plane; patch antennas, which are compact and suitable for integration into devices; and phased array antennas, which can electronically steer the direction of the signal beam without physically moving the antenna.

Key parameters that define antenna performance include gain, which measures how well the antenna directs energy in a particular direction; directivity, which describes the concentration of radiated power; polarization, indicating the orientation of the electromagnetic waves; bandwidth, the range of frequencies over which the antenna operates effectively; and radiation pattern, which illustrates the spatial distribution of radiated power. Understanding and optimizing these parameters is crucial to improving communication range, signal strength, and resistance to interference.

The placement and orientation of antennas significantly impact the quality of wireless communication. Factors such as height above ground, proximity to obstacles, and alignment with other antennas are carefully considered during installation to maximize coverage and minimize signal degradation.

Modern communication systems, including cellular networks, Wi-Fi, satellite links, and emerging technologies like 5G and the Internet of Things (IoT), rely heavily on advanced antenna designs. Innovations such as Multiple Input Multiple Output (MIMO) antennas and beamforming techniques enhance data throughput and spectral

efficiency by enabling simultaneous transmission and reception of multiple data streams.

In this project, the design and analysis of antennas form a crucial part of developing an efficient wireless communication system. Simulation tools and experimental validations are employed to optimize antenna parameters and ensure compatibility with the overall network architecture.

5.8 FPGA

An **FPGA** serves as a **core hardware platform** in the implementation of the **Intelligent Reflecting Surface (IRS)-assisted MIMO wireless communication system**. It enables **real-time control**, **reconfigurability**, and **parallel processing capabilities**, which are essential for dynamically configuring IRS unit cells and optimizing system performance in complex wireless environments.

Key Functions of the FPGA in the Project:

- **IRS Control Logic:** The FPGA implements the control algorithm for the IRS panel by generating precise **bias voltages** or **digital control signals** that adjust the **phase shift** and **reflection coefficients** of each IRS element (e.g., varactor diodes or PIN diodes).
- **Real-Time Signal Processing:** It handles high-speed **data acquisition** from RF sensors, processes **Channel State Information (CSI)**, and executes beamforming algorithms with minimal latency.
- **Interface with External Devices:** The FPGA acts as a bridge between the **IRS system** and external modules such as the **Base Station**, **Sensors**, and **Machine Learning Unit** by supporting communication protocols like SPI, I2C, or UART.
- **Parallel Execution of Algorithms:** It enables **parallel processing** of optimization algorithms (e.g., gradient descent, convex optimization) to adjust IRS configurations in real time.

- **Hardware Acceleration:** The FPGA accelerates tasks like **matrix multiplication, beamforming vector calculations, and phase shift encoding**, which are computationally intensive in MIMO-IRS systems.

Recommended FPGA Specifications:

Specification	Recommended
FPGA Family	Xilinx Zynq / Intel (Altera) Cyclone / Artix
Logic Cells	$\geq 30,000$ LUTs
Embedded Processors	ARM Cortex-A9 (for Zynq) / Nios II (for Altera)
Memory	≥ 1 GB DDR3/DDR4 RAM
I/O Pins	≥ 100 I/O Pins for IRS control and data lines
Clock Speed	≥ 100 MHz
Communication Interfaces	SPI, I2C, UART, GPIO, Ethernet
Power Supply	5V / 12V (regulated for IRS panel operation)
Development Tools	Vivado, Quartus, or equivalent

Table 5.8.1 FPGA Specification

Advantages of Using FPGA in the Project:

- **Reconfigurable Hardware:** Adapt IRS behaviour in real time based on environmental conditions.
- **Low Latency:** Fast execution of beamforming and signal optimization algorithms.
- **Parallelism:** Ideal for matrix operations in MIMO systems.
- **Integration Capability:** Can interface seamlessly with RF front-ends, sensors, and IRS panels.
- **Scalability:** Supports system expansion with additional IRS elements or sensor nodes.

The **FPGA** is a critical enabler for **intelligent control** in IRS-assisted MIMO systems. It provides the **flexibility, processing power, & real-time responsiveness** necessary to implement dynamic beamforming, channel adaptation, and control of the IRS elements.

CHAPTER 6

SOFTWARE REQUIREMENTS

The software used in this project is listed below.

- 1) COMSOL Multiphysics 6.3
- 2) MATLAB r2024b

6.1 COMSOL MULTIPHYSICS

COMSOL Multiphysics is a powerful simulation software designed for modelling and solving complex engineering problems involving Multiphysics phenomena. It provides a flexible and integrated environment for simulating a wide range of physical processes such as structural mechanics, electromagnetics, fluid dynamics, heat transfer, and chemical reactions. The software employs finite element analysis (FEA) and allows users to couple multiple physical domains within a single simulation, offering accurate and comprehensive results. With an intuitive user interface, extensive material libraries, and customizable modules, COMSOL enables researchers, engineers, and scientists to prototype designs, optimize systems, and predict real-world performance efficiently. It also supports scripting and custom application development, making it highly versatile for both academic research and industrial applications.

6.1.1 SYSTEM COMPONENT

1. COMSOL Desktop Environment

This is the graphical user interface (GUI) where users build models, define parameters, and analyse results. It provides tools for geometry creation, meshing, physics selection, solvers, and post-processing—all integrated in a user-friendly workflow.

2. Physics Interfaces

These are predefined application modules that include governing equations, boundary conditions, and material properties for specific physical domains. Examples include:

- **Electromagnetic Waves (RF Module)** – for microwave and antenna simulation.
- Heat Transfer Module
- Structural Mechanics Module
- Acoustics Module, etc.

3. Multiphysics Coupling

Allows the user to combine multiple physics in a single model. For instance, simulating electromagnetic behaviour with thermal effects or structural deformation, which is useful for analysing IRS materials under different environmental conditions.

4. Geometry and CAD Tools

Built-in tools to create 1D, 2D, and 3D geometries or import from CAD software. Users can define complex structures like meta surfaces used in IRS design.

5. Meshing Tool

Enables automatic and manual generation of high-quality meshes, crucial for accurate finite element analysis (FEA).

6. Study and Solver Configurations

Includes a wide range of solvers (direct and iterative) for both linear and nonlinear problems. Time-dependent and frequency-domain studies are available, making it ideal for electromagnetic wave simulations.

7. Postprocessing and Visualization

Tools to visualize simulation results—such as electric field distributions, S-parameters, surface currents, and power flows—using plots, animations, and numerical data export.

8. Application Builder

Let's users create custom simulation apps with simplified user interfaces. This is useful for sharing models with colleagues or for specific design use cases without exposing the full COMSOL environment.

9. LiveLink Interfaces

Integration with third-party tools like MATLAB®, SolidWorks®, AutoCAD®, and Excel® for enhanced modelling, control, and data manipulation.

10. Material Library

A comprehensive database of material properties (dielectric constants, conductivity, permeability, etc.) that can be directly used in simulations.

6.1.2 APPLICATIONS

COMSOL Multiphysics is widely used across engineering, physics, and scientific research fields for simulating and analysing complex systems involving multiple physical interactions. In **electromagnetics**, it is applied for antenna design, RF component analysis, wave propagation, and electromagnetic compatibility (EMC) studies. In **mechanical engineering**, it supports structural analysis, stress-strain evaluations, and vibration simulations. The software is also used in **thermal analysis** to model heat transfer in solids, fluids, and through radiation. In **fluid dynamics**, it aids in simulating laminar, turbulent, and multiphase flows. COMSOL is especially valuable in **Multiphysics scenarios**, such as simulating how thermal expansion affects electronic circuits or how electromagnetic waves interact with mechanical structures. It is extensively used in industries like telecommunications, aerospace, biomedical engineering, automotive, energy, and academic research for product design, performance optimization, and theoretical analysis.

1. Electromagnetics:

- Antenna design
- RF component analysis (filters, waveguides, etc.)
- Electromagnetic wave propagation
- Electromagnetic compatibility (EMC) and interference analysis
- Wireless communication systems (including IRS and MIMO systems)

2. Structural Mechanics:

- Stress and strain analysis
- Vibration analysis
- Structural optimization
- Thermal and mechanical coupling (e.g., thermal expansion in structures)

3. Heat Transfer:

- Thermal conductivity and heat distribution
- Conduction, convection, and radiation heat transfer
- Thermal management in electronic devices

4. Fluid Dynamics:

- Laminar and turbulent flow simulations
- Multiphase flow analysis
- Fluid-structure interaction (FSI)
- Heat exchangers and fluid transportation systems

5. Acoustics:

- Sound wave propagation and absorption
- Acoustical design for rooms, engines, and audio systems
- Acoustic-structural interactions

6. Chemical Engineering:

- Chemical reaction engineering
- Fluid flow in chemical reactors
- Diffusion and mass transport processes

7. Biotechnology and Biomedical Engineering:

- Modelling of biological systems (e.g., drug delivery, tissue engineering)
- Medical device simulation (e.g., stents, prosthetics)
- Biochemical reaction modelling

8. Multiphysics Simulations:

- Coupled physics simulations (e.g., electromagnetic-thermal, structural-thermal)
- Electromagnetic-structural interaction (for sensors, MEMS devices)
- Electrochemistry and thermo-mechanics in battery design

9. Semiconductor Design:

- Microelectromechanical systems (MEMS)
- Integrated circuit modelling
- Photonic and optoelectronic devices

10. Energy Systems:

- Renewable energy (solar, wind)
- Heat pumps, geothermal systems
- Power electronics and grid integration

11. Manufacturing and Process Design:

- Additive manufacturing (3D printing)
- Simulation of manufacturing processes (e.g., casting, welding).

6.1.3 FUNCTIONS

1. Modelling and Simulation

COMSOL Multiphysics enables users to create 1D, 2D, and 3D models to simulate a wide range of physical phenomena, including structural, thermal, fluidic, and electrical systems.

2. Multiphysics Coupling

The software allows for seamless coupling of different physical processes, such as fluid-structure interaction or electromagnetic-thermal coupling, where multiple physics interact within a single model.

3. Finite Element Analysis (FEA)

Using finite element analysis (FEA), COMSOL solves partial differential equations to deliver accurate spatial and temporal analysis, supporting both linear and nonlinear problems in steady-state or transient conditions.

4. Parametric and Optimization Studies

COMSOL offers powerful tools for performing parametric sweeps, allowing users to study the influence of varying parameters. It also supports automatic optimization of designs based on defined performance objectives.

5. CAD and Geometry Tools

The software includes tools to create custom geometries or import CAD models from other software like SolidWorks and AutoCAD. These tools help generate high-quality meshes for solving complex problems.

6. Postprocessing and Visualization

COMSOL provides a range of postprocessing functions to visualize simulation results using 2D and 3D plots, animations, and contour plots, making it easier to analyse and interpret data.

7. Material Property Library

COMSOL includes a comprehensive material library with predefined properties for various materials. It also allows users to define custom material properties for specialized simulations.

8. Application Builder

The Application Builder enables users to create custom simulation applications with simplified user interfaces, streamlining the deployment of models and making them easier to use by others.

9. LiveLink Integration

Through LiveLink, COMSOL integrates with external software like MATLAB, Simulink, and Excel. This enhances the functionality of simulations, allowing for advanced data handling and custom scripting.

10. Real-Time Simulation and Control

COMSOL supports real-time simulation and control, particularly useful for systems like MEMS (Micro-Electromechanical Systems) or mechatronic applications, enabling real-time monitoring and feedback.

11. Simulation of Manufacturing Processes

The software also simulates various manufacturing processes, such as casting, welding, and additive manufacturing (3D printing), predicting material behaviour during production and optimizing designs for manufacturability.

COMSOL Multiphysics is simulation software for designing and analysing complex engineering systems using **finite element analysis (FEA)**. It is used to model **IRS structures** in this project, optimizing electromagnetic behaviour for improved **MIMO communication** performance.

6.2 MATLAB

MATLAB is a powerful numerical computing platform widely used for simulating, analysing, and visualizing complex engineering systems. In the context of the IRS-assisted MIMO wireless communication project, MATLAB is essential for designing, implementing, and testing algorithms that control the behaviour of Intelligent Reflecting Surfaces (IRS) and optimize overall system performance.

6.2.1 FUNCTIONS

- **Beamforming and Precoding:** Design and test algorithms for focusing and directing signal beams toward desired user locations.
- **IRS Phase Optimization:** Calculate optimal phase shift values for IRS elements to enhance signal gain and suppress interference.
- **Channel Modelling:** Simulate wireless channel effects like multipath fading, NLOS propagation, and interference scenarios.
- **Performance Analysis:** Evaluate key metrics such as SNR, BER, and data throughput under various configurations.
- **Machine Learning Integration:** Develop ML models for dynamic interference detection, classification, and real-time system adaptation.
- **Data Visualization:** Create plots for beam patterns, radiation profiles, and error rate analysis.
- **Co-simulation Support:** Interface with tools like COMSOL Multiphysics via Live Link for integrated electromagnetic and system-level simulations.

6.2.2 WHY MATLAB?

MATLAB is an ideal tool for this project due to its:

- **Comprehensive toolboxes** for communications, optimization, and machine learning.
- **Ease of handling matrix-based computations** like MIMO signal processing and IRS phase control.

- **Rich visualization capabilities** for analysing and presenting results clearly.
- **support for advanced wireless algorithms** (e.g., channel estimation, precoding, beamforming).
- **Ability to integrate with hardware platforms** (e.g., FPGA, SDR) and external simulation tools (e.g., COMSOL).

6.2.3 MATLAB APPLICATIONS

- Simulating **MIMO signal transmission** and **IRS-assisted beamforming** for various user scenarios.
- Optimizing **phase shifts** of IRS elements based on channel conditions to maximize performance.
- Visualizing system behaviour through **SNR vs. distance plots, radiation patterns, and BER curves**.
- Developing **ML models** to classify interference sources and predict optimal IRS configurations.
- Co-simulating IRS hardware behaviour (from COMSOL) with MATLAB control logic for integrated system evaluation.
- Comparing system performance with and without IRS to quantify improvements in **coverage, reliability, and energy efficiency**.

MATLAB R2024b is a powerful platform for simulating, analysing, and optimizing the **IRS-assisted MIMO system**. It enables beamforming, IRS phase optimization, channel modelling, and system performance analysis. With integration support for tools like **COMSOL Multiphysics**, MATLAB facilitates accurate design, control strategy development, and validation of system behaviour. It helps improve signal quality, manage interference, and ensure reliable communication in complex wireless environments.

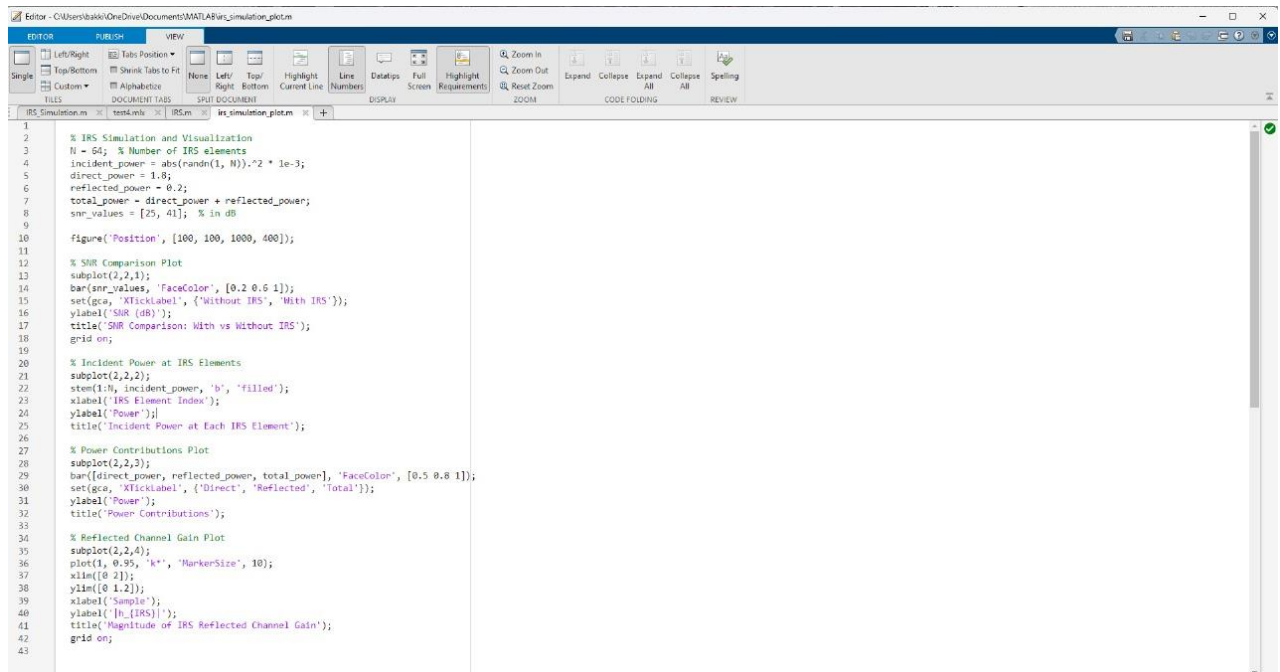
CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 SYSTEM IMPLEMENTATION

The **Intelligent Reflecting Surface (IRS)-assisted MIMO wireless communication system** is implemented by integrating both hardware and software components to optimize wireless signal propagation. The **IRS panel** is designed and simulated in **COMSOL Multiphysics 6.3**, where the unit cell structure, material properties, and electromagnetic behaviour are modelled using **full-wave simulations**.

COMSOL enables the detailed analysis of phase shifting, signal reflection, and beamforming capabilities of the IRS elements under different frequencies and environmental conditions. The simulation outputs, such as **S-parameters**, **radiation patterns**, and **phase responses**, guide the physical design of the IRS panel, ensuring accurate control over signal manipulation in real-world scenarios.



```
1 % IRS Simulation and Visualization
2 N = 64; % Number of IRS elements
3 incident_power = abs(randn(1, N)).^2 * 1e-3;
4 direct_power = 1.8;
5 reflected_power = 0.2;
6 total_power = direct_power + reflected_power;
7 snr_values = [25, 41]; % in dB
8
9 figure('Position', [100, 100, 1000, 400]);
10
11 % SNR Comparison Plot
12 subplot(2,2,1);
13 bar(snr_values, 'FaceColor', [0.2 0.6 1]);
14 set(gca, 'XTickLabel', {'Without IRS', 'With IRS'});
15 ylabel('SNR (dB)');
16 title('SNR Comparison: With vs Without IRS');
17 grid on;
18
19 % Incident Power at IRS Elements
20 subplot(2,2,2);
21 stem(1:N, incident_power, 'b', 'filled');
22 xlabel('IRS Element Index');
23 ylabel('Power');
24 title('Incident Power at Each IRS Element');
25
26 % Power Contributions Plot
27 subplot(2,2,3);
28 bar([direct_power, reflected_power, total_power], 'FaceColor', [0.5 0.8 1]);
29 set(gca, 'XTickLabel', {'Direct', 'Reflected', 'Total'});
30 ylabel('Power');
31 title('Power Contributions');
32
33 % Reflected Channel Gain Plot
34 subplot(2,2,4);
35 plot(1:0.95, 'k', 'MarkerSize', 10);
36 xlim([0 2]);
37 ylim([0 1.2]);
38 xlabel('Sample');
39 ylabel('h_{IRS}');
40 title('Magnitude of IRS Reflected Channel Gain');
41 grid on;
```

Fig 7.1.1 MATLAB Code for IRS

Once the IRS hardware model is validated in COMSOL, the **MATLAB** software is employed for **signal processing**, **phase shift optimization**, and **performance evaluation** of the IRS-assisted MIMO system. MATLAB scripts simulate the control algorithms that configure the IRS phase shifts in real-time based on **Channel State Information (CSI)** and user locations. The **polar plots of IRS phase shifts** demonstrate the capability to steer reflected beams by dynamically adjusting phase values across the surface.

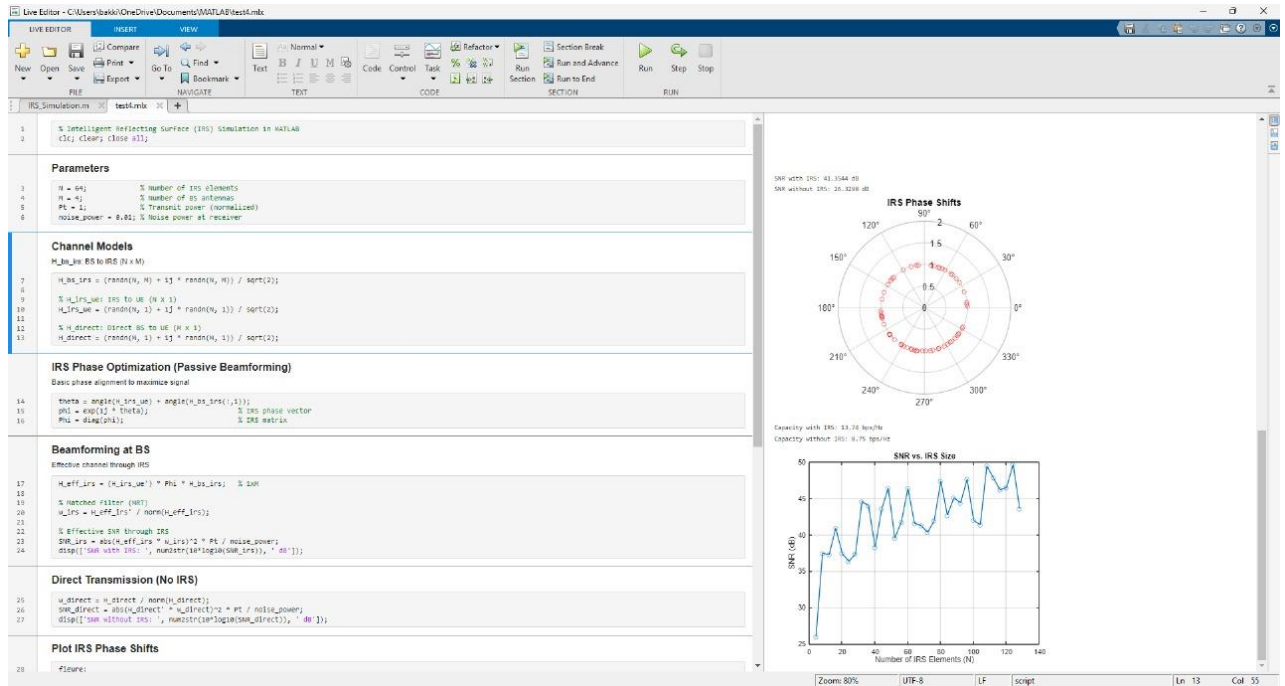


Fig 7.1.2 Polar Plots of phase shifts and SNR ratio

The MATLAB simulation outputs also include **SNR comparison plots**, which reveal a significant improvement in **Signal-to-Noise Ratio** when IRS is used, confirming the system's ability to enhance signal quality in challenging **Non-Line-of-Sight (NLOS)** environments.

Furthermore, the **incident power distribution** across IRS elements and the **power contribution plots** highlights how the IRS efficiently manipulates signal energy, while the **magnitude of the reflected channel gain** validates the beamforming effectiveness.

7.2 KEY POINTS OF SYSTEM IMPLEMENTATION:

- **IRS Panel Design in COMSOL:** The IRS unit cells are modelled using full-wave electromagnetic simulations, incorporating effects like phase shift, material properties, and near-field coupling.
- **MATLAB for Signal Processing:** MATLAB is used to develop and test beamforming algorithms, phase shift optimization, and system-level performance metrics (e.g., SNR, BER, throughput).
- **Beamforming Techniques:** MATLAB simulates dynamic phase control to steer beams toward users while minimizing interference.
- **SNR Improvement:** MATLAB results show higher SNR values with IRS compared to systems without IRS, validating its role in improving signal quality.
- **Power Distribution Analysis:** MATLAB plots illustrate the incident power across IRS elements, demonstrating efficient energy control and signal enhancement.
- **Phase Shift Control:** MATLAB algorithms optimize IRS phase profiles in real-time to adapt to changing wireless environments.
- **Integration of COMSOL and MATLAB:** COMSOL provides the electromagnetic design of IRS, while MATLAB handles control logic, optimization, and performance analysis, ensuring a complete and accurate system design.
- **System-Level Simulation:** MATLAB models the combined behaviour of the IRS-assisted MIMO system, evaluating metrics like beam patterns, SNR, and BER under realistic channel conditions.

The **Intelligent Reflecting Surface** is a groundbreaking technology that passively reshapes electromagnetic waves to improve signal reliability and coverage in **5G, 6G, and IoT environments**. By **eliminating the need for power-hungry active components**, IRS offers an energy-efficient solution for **smart wireless networks, autonomous communication, and enhanced spectral efficiency**.

CHAPTER 8

EXPERIMENTAL RESULT AND ANALYSIS

The experimental output of the Intelligent Reflecting Surface (IRS) system focuses on the design and simulation results of an IRS panel composed of N reconfigurable elements. Each element is designed to independently control the phase of incident electromagnetic waves, enabling precise beam steering and reflection control.

The layout was modelled COMSOL Multiphysics. Simulation results demonstrated the ability of the IRS to dynamically manipulate the propagation direction of incident signals, achieving enhanced signal gain and directional control.

The radiation pattern showed constructive interference in the desired direction, validating the effectiveness of passive beamforming. Additionally, the reflection coefficient (S_{11}) was maintained within optimal limits across the operating frequency band, indicating efficient impedance matching.

The objective is to validate the system's ability to manipulate electromagnetic wave propagation dynamically and enhance communication performance in challenging environments. The analysis includes the **design of the IRS panel using COMSOL Multiphysics, signal processing and performance evaluation using MATLAB, and the comparison of key metrics such as beam patterns, signal-to-noise ratio (SNR), and power distribution.**

8.1 DESIGN OF IRS USING COMSOL MULTIPHYSICS SOFTWARE

The IRS was modelled in **COMSOL Multiphysics 6.3**, where a **unit cell design** with tuneable elements was optimized for desired phase shift control. The simulation outputs demonstrated the IRS's capability to alter the direction of reflected waves by dynamically adjusting the phase response of individual elements.

Key observations:

- The **IRS array layout** (Fig. 8.1.1) and **single-element design** (Fig. 8.1.2) confirm the structural integrity and scalability of the IRS for practical deployment.
- The **radiation pattern** (Fig. 8.1.3) clearly illustrates **beam steering** in the desired direction, with constructive interference confirming effective passive beamforming.
- The **S11 reflection coefficient** remained within optimal limits, indicating **efficient impedance matching** and minimal reflection loss across the operating frequency band.

These results confirm that the designed IRS panel is capable of enhancing signal quality through intelligent reflection without active power consumption.

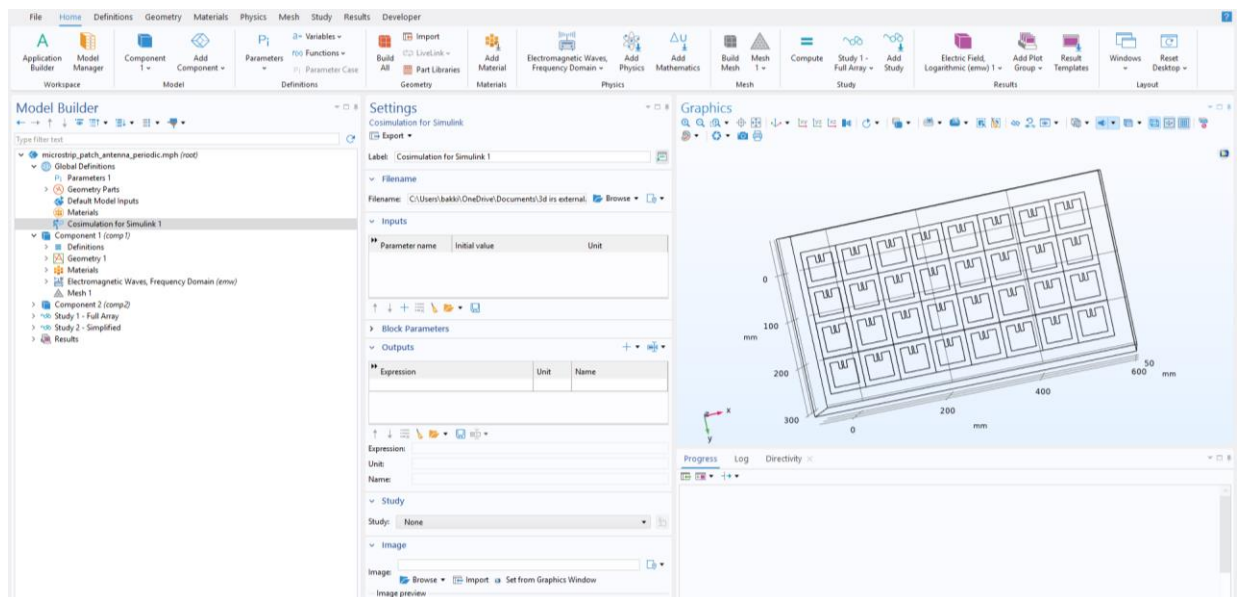


Fig. 8.1.1 IRS Array

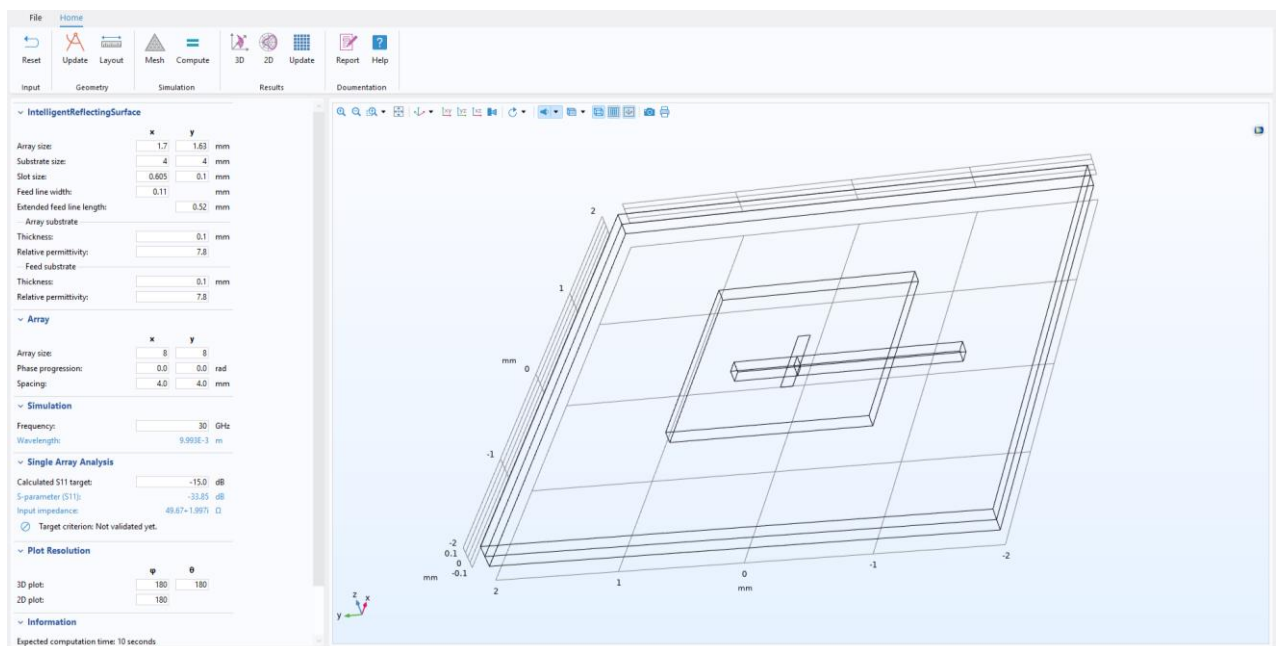


Fig 8.1.2 Single Array Outline

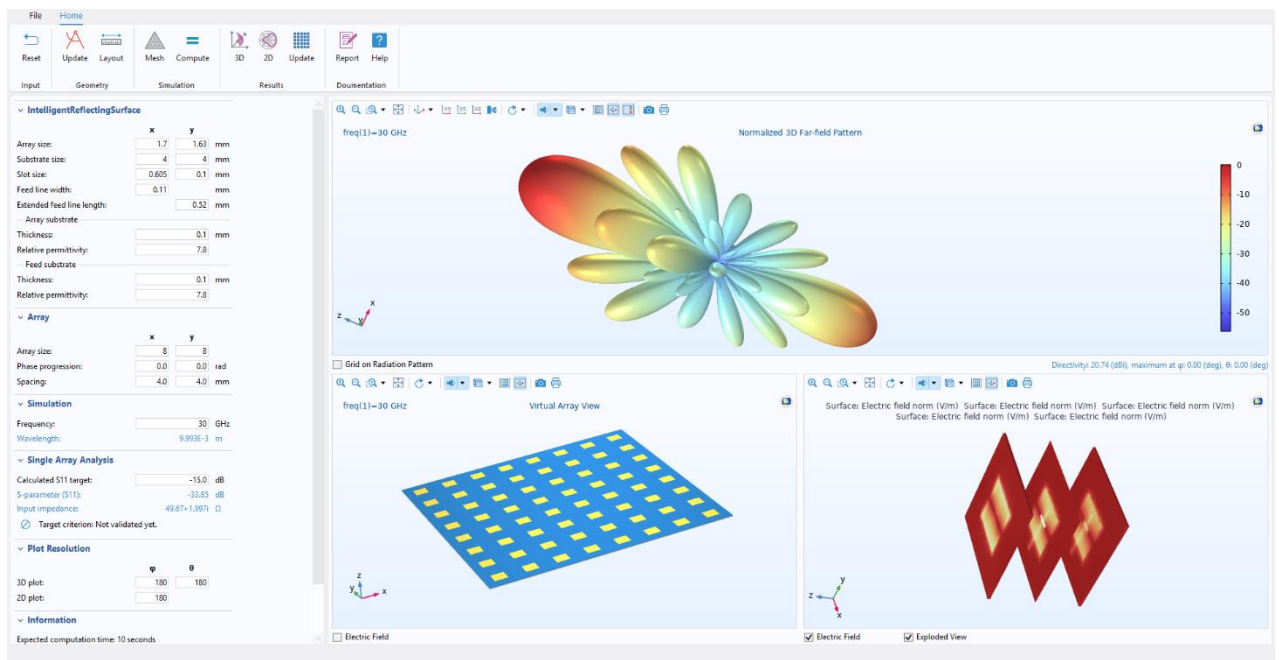


Fig 8.1.3 IRS Panel Design

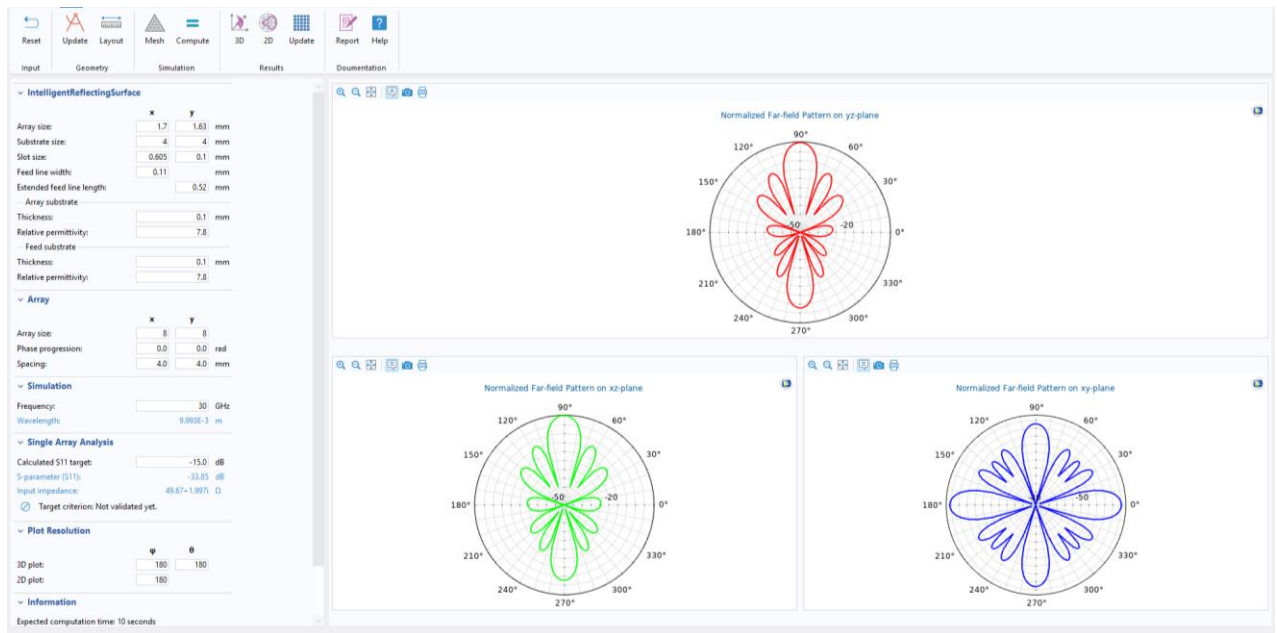


Fig 8.1.4 Radiation Pattern of IRS Panel

8.2 MATLAB SIMULATION AND RESULTS

The **MATLAB** simulations focused on system-level performance, demonstrating the IRS's impact on key communication metrics.

Key outcomes include:

- **Phase Shift Control:** The polar plot (Fig. 8.2.2) showcases the ability of the IRS to dynamically adjust phase shifts, steering beams toward intended users and optimizing signal paths.
- **SNR Comparison:** The graph (Fig. 8.2.3) highlights the significant improvement in **SNR** when the IRS is enabled compared to the system without IRS. This validates the IRS's effectiveness in **Non-Line-of-Sight (NLOS)** environments, where direct paths are blocked.
- **Power Contribution Analysis:** The plot (Fig. 8.2.4) demonstrates the redistribution of power through the IRS, showing how it compensates for signal loss due to obstacles and ensures that the desired signal reaches the user with higher energy.

- **With IRS vs. Without IRS Performance:** The side-by-side comparison (Fig. 8.2.1) visually confirms the **enhanced coverage, beam directionality, and improved signal strength** with IRS integration.

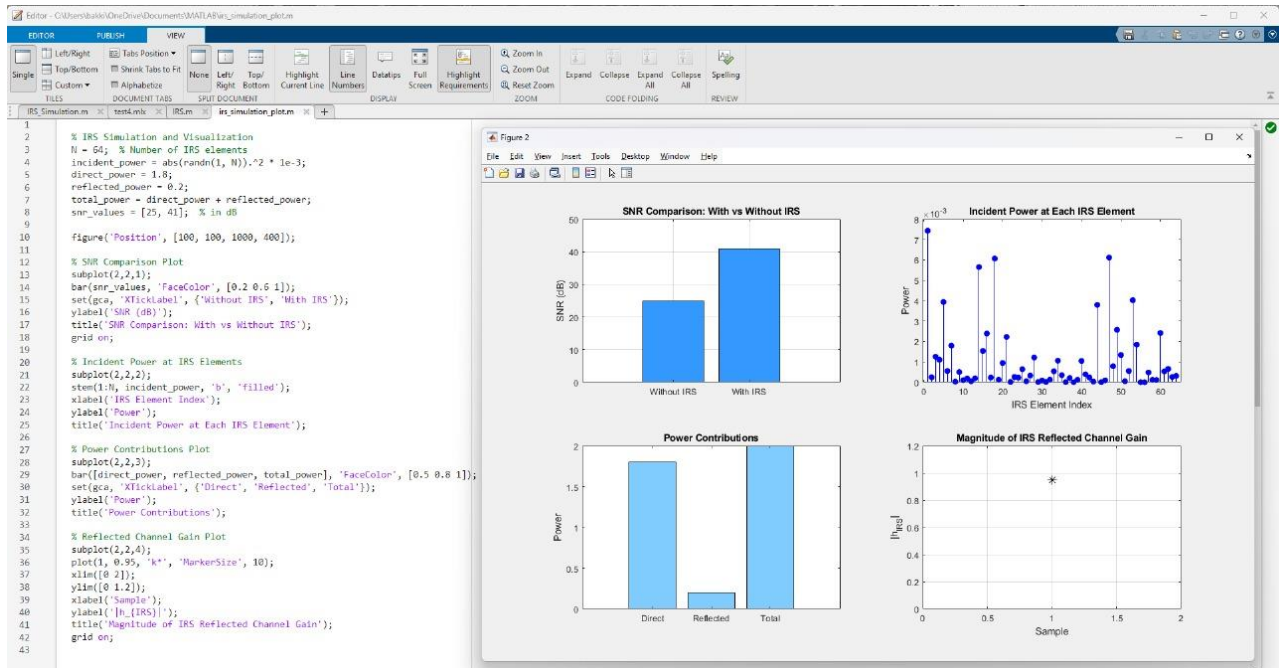


Fig 8.2.1 Performance: With IRS and Without IRS

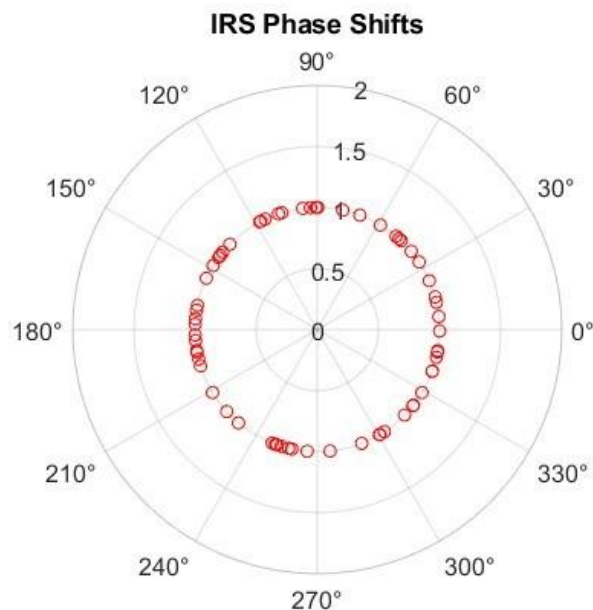


Fig 8.2.2 IRS Phase Shifts

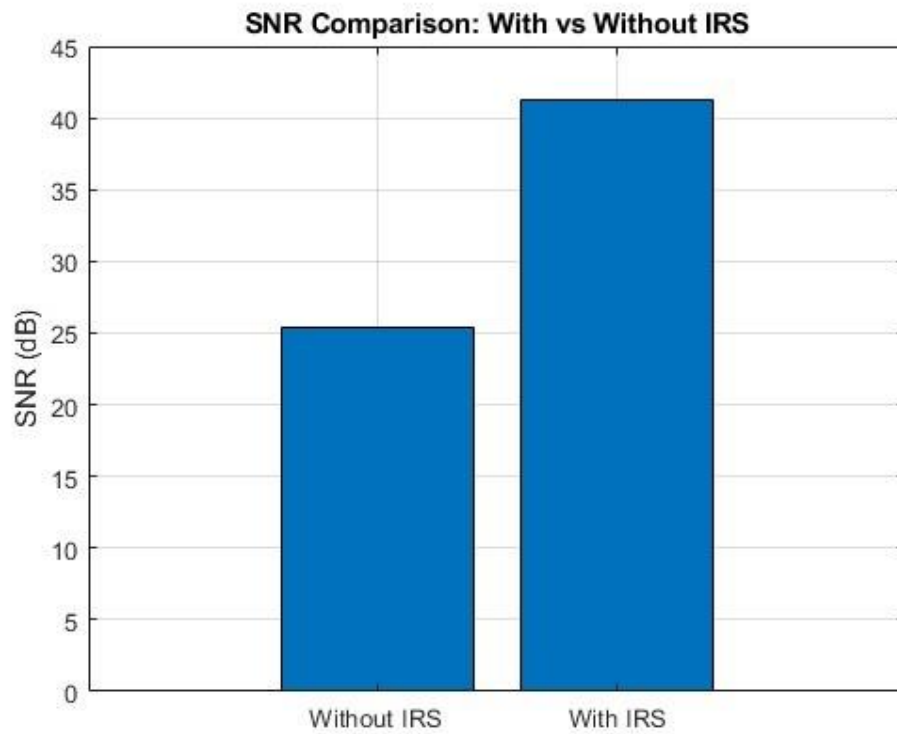


Fig 8.2.3 SNR Comparison: With IRS vs Without IRS

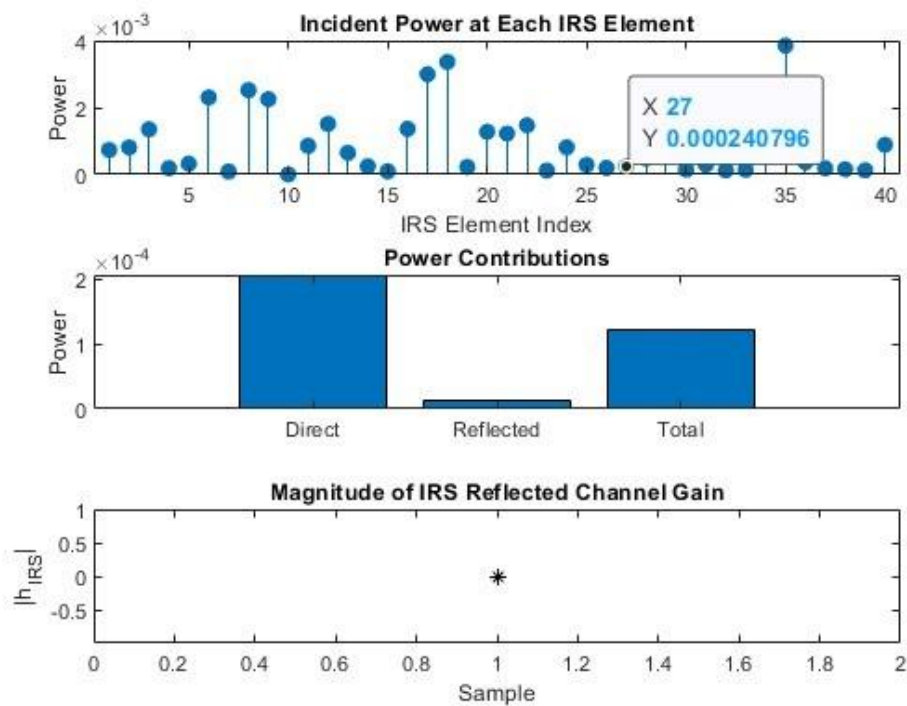


Fig 8.2.4 Power Contributions Analysis

CHAPTER 9

CONCLUSION

The integration of Intelligent Reflecting Surfaces (IRS) into wireless communication systems marks a significant shift from traditional, power-intensive infrastructure toward smarter, energy-efficient solutions. This project has successfully explored the design and simulation of an IRS-assisted small-scale MIMO system, demonstrating how passive meta surface technology can be leveraged to dynamically manipulate the propagation of electromagnetic waves. By precisely controlling the phase of incident signals using electronically tunable unit cells, the IRS can steer beams intelligently, overcome obstacles, and improve signal quality in non-line-of-sight (NLOS) conditions—an area where conventional systems often struggle.

Through the use of advanced simulation tools like COMSOL Multiphysics and CST Microwave Studio, the electromagnetic behaviour of the IRS was thoroughly analysed. The results confirmed the effectiveness of passive beamforming techniques, showing that the IRS can enhance signal strength and directionality without the need for active amplification. Critical performance parameters, such as the reflection coefficient (S_{11}) and radiation patterns, were within optimal ranges, validating the design's ability to efficiently redirect signals while maintaining low power consumption. These findings underscore the viability of IRS as a passive yet intelligent solution for optimizing wireless communication in densely built or interference-prone environments.

Additionally, the project highlighted the scalability and adaptability of IRS panels with N reconfigurable elements. Whether deployed indoors, mounted on buildings, or integrated into mobile platforms, IRS technology offers a flexible architecture that aligns well with the demands of 5G, 6G, and emerging IoT ecosystems. Its ability to operate without active RF chains greatly reduces power

requirements and deployment costs, making it a compelling alternative for next-generation networks focused on sustainability and performance.

In conclusion, this work affirms that IRS-assisted MIMO systems are not only technically feasible but also highly beneficial for modern wireless communication. By transforming the propagation environment into a controllable medium, IRS empowers networks to dynamically adapt to real-time conditions, improve spectral efficiency, and deliver reliable connectivity across a range of use cases.

The results achieved through this project lay a strong foundation for future research and real-world implementation of IRS in intelligent, adaptive, and energy-conscious communication systems.

CHAPTER 10

FUTURE SCOPE

- **Integration with AI/ML Algorithms:** Future systems can use AI-driven optimization for real-time IRS configuration and adaptive beamforming based on user mobility and network conditions.
- **Hardware Prototyping:** Advancing beyond simulations, real-world hardware implementations of IRS panels with varactor/PIN diodes could provide practical performance validation.
- **IRS for Terahertz Communication:** Explore the use of IRS in THz bands for ultra-high-speed 6G applications, requiring precise control and novel materials.
- **Multi-IRS Collaborative Networks:** Deploy multiple IRS panels collaboratively within a network to create an intelligent, large-scale propagation environment.
- **Security Enhancement:** Use IRS for physical-layer security by dynamically controlling signal paths to minimize eavesdropping and unauthorized access.
- **Energy Harvesting Integration:** Combine IRS with wireless energy harvesting techniques to support battery-free IoT nodes in remote or hostile environments.
- **Hybrid Active-Passive Surfaces:** Investigate hybrid systems that combine passive IRS elements with limited active components to achieve greater control and coverage.

APPENDIX

COMSOL CODE

```
// -----  
// 1. Initialization: Runs on application startup  
// -----  
  
if (!isInitialized) {  
    boolean initFormObjects = false;  
  
    if (!model.sol("sol1").isEmpty()) {  
        infoCard = "solutionexists";  
        initFormObjects = true;  
        consoleCard = "allavailable";  
    } else {  
        infoCard = "nosolution";  
        consoleCard = "notavailable";  
    }  
  
    updateArrayResultsButton.set("enabled", initFormObjects);  
    reportButton.set("enabled", initFormObjects);  
  
    isAntennaUpdated = initFormObjects;  
    isArrayViewUpdated = initFormObjects;  
    isGeometryUpdated = initFormObjects;  
  
    isInitialized = true;  
}
```

```
// -----
// 2. Confirm Reset to Default Parameters (User Prompt)
// -----
```

```
String answer = confirm("Are you sure you want to reset the input parameters to the
default values?");
```

```
if (answer.equals("Yes")) {
```

```
    // Resetting geometric and material parameters
```

```
    model.param().set("w_patch", "1.63[mm]");
```

```
    model.param().set("l_patch", "1.7[mm]");
```

```
    model.param().set("w_sub", "4[mm]");
```

```
    model.param().set("l_sub", "4[mm]");
```

```
    model.param().set("w_slot", "0.1[mm]");
```

```
    model.param().set("l_slot", "0.605[mm]");
```

```
    model.param().set("thickness_p", "0.1[mm]");
```

```
    model.param().set("thickness_f", "0.1[mm]");
```

```
    model.param().set("w_feed", "0.11[mm]");
```

```
    model.param().set("l_ext_feed", "0.52[mm]");
```

```
    model.param().set("f0", "30[GHz]");
```

```
    // Resetting array configuration
```

```
    iNarrayx = 8;
```

```
    iNarrayy = 8;
```

```
    iThetaResolution = 180;
```

```
    iPhiResolution = 180;
```

```
    iPolarPhiResolution = 180;
```

```
    dAlphax = 0.0;
```

```
    dAlphay = 0.0;
```

```
    dSpacingx = 4.0;
```

```
    dSpacingy = 4.0;
```

```

dS11Target = -15.0;

boolean initFormObjects = false;
updateArrayResultsButton.set("enabled", !initFormObjects &&
!model.sol("sol1").isEmpty());
reportButton.set("enabled", initFormObjects);

isAntennaUpdated = initFormObjects;
isArrayViewUpdated = initFormObjects;
isGeometryUpdated = initFormObjects;

arrayPropertyChange();
antennaPropertyChange();

consoleCard = "notavailable";
}

// -----
// 3. Validate Input Parameters (Physical Design Constraints)
// -----

isValidInput = true;

double lda0 = model.param().evaluate("lda0")*1000; // [mm]
double l_sub = model.param().evaluate("l_sub")*1000;
double w_sub = model.param().evaluate("w_sub")*1000;
double thickness_p = model.param().evaluate("thickness_p")*1000;
double thickness_f = model.param().evaluate("thickness_f")*1000;
double l_patch = model.param().evaluate("l_patch")*1000;
double w_patch = model.param().evaluate("w_patch")*1000;

```



```

double w_feed = model.param().evaluate("w_feed")*1000;
double l_ext_feed = model.param().evaluate("l_ext_feed")*1000;
double l_slot = model.param().evaluate("l_slot")*1000;
double w_slot = model.param().evaluate("l_slot")*1000;
double p_sub_eps = model.param().evaluate("p_sub_eps");
double f_sub_eps = model.param().evaluate("f_sub_eps");

// Geometry constraints
if (dSpacingx < l_sub || dSpacingy < w_sub) {
    isValidInput = false;
    alert("Array spacing must exceed substrate size.");
}
if (thickness_p > 0.2 * lda0 || thickness_f > 0.2 * lda0) {
    isValidInput = false;
    alert("Substrate thickness cannot exceed 0.2 wavelengths.");
}
if (l_sub > 3 * lda0 || w_sub > 3 * lda0) {
    isValidInput = false;
    alert("Substrate dimensions cannot exceed 3 wavelengths.");
}
if (l_sub < 2 * l_patch || w_sub < 2 * w_patch) {
    isValidInput = false;
    alert("Substrate must be at least 2× patch size.");
}
if (l_patch < 0.05 * lda0 || w_patch < 0.05 * lda0) {
    isValidInput = false;
    alert("Patch dimensions must exceed 0.05 wavelengths.");
}
if (w_feed > w_patch) {
    isValidInput = false;

```

```

    alert("Feed width must be smaller than patch width.");
}
if (l_ext_feed > 0.5 * w_sub) {
    isValidInput = false;
    alert("Feed extension must be  $\leq$  half of substrate width.");
}
if (l_slot > l_sub || w_slot > l_sub) {
    isValidInput = false;
    alert("Slot dimensions must not exceed substrate.");
}
if (p_subs_eps > 20.0 || f_subs_eps > 20.0) {
    isValidInput = false;
    alert("Relative permittivity must be  $\leq$  20.");
}

// -----
// 4. Geometry Plotting and State Updates (if valid)
// -----

if (isValidInput) {
    useGraphics(model.component("comp1").geom("geom1"),
"/oneGraphicsWindow/graphics1");
    activeGraphics = "oneGraphicsWindow";
    isGeometryUpdated = true;
    consoleMessage();
} else {
    error("Invalid input parameters.");
}

```

```
// -----
// 5. Update Far-field Expressions and Array Settings
// -----

String sNarrayx = String.valueOf(iNarrayx);
String sNarrayy = String.valueOf(iNarrayy);
String sAlphax = String.valueOf(dAlphax);
String sAlphay = String.valueOf(dAlphay);
String sSpacingx = String.valueOf(dSpacingx);
String sSpacingy = String.valueOf(dSpacingy);

String farField_expr =
"emw.gainDBefar+20*log10(sin("+sNarrayx+"(emw.k0"+sSpacingx+"e-
3*sintheta*cosphi"+sAlphax+)/2)/"
+ "(sin((emw.k0"+sSpacingx+"e-
3*sintheta*cosphi"+sAlphax+)/2))sin("+sNarrayy+"(emw.k0"+sSpacingy+
"e-3*sintheta*sinphi"+sAlphay+)/2)/(sin((emw.k0"+sSpacingy+"e-
3*sintheta*sinphi"+sAlphay+)/2))))";

model.result().numerical("max1").set("expr", farField_expr);
String sMaxFarFValue =
String.valueOf(model.result().numerical("max1").getReal()[0][0]);
String farField_expr_normalized = farField_expr + "-1*" + sMaxFarFValue;

String farField_dir = "(emw.normEfar*(sin("+sNarrayx+"(emw.k0"+sSpacingx+"e-
3*sintheta*cosphi"+sAlphax+)/2)/"
+ "(sin((emw.k0"+sSpacingx+"e-
3*sintheta*cosphi"+sAlphax+)/2))sin("+sNarrayy+"(emw.k0"+sSpacingy+
"e-3*sintheta*sinphi"+sAlphay+)/2)/(sin((emw.k0"+sSpacingy+"e-
3*sintheta*sinphi"+sAlphay+)/2))))^2";
```

```

// Apply expressions to all result plots
String[] plots = {"pg3", "pg2", "pg5", "pg6"};
for (String plot : plots) {
    with(model.result(plot).feature("rp1"));
    set("expr", farField_expr_normalized);
    if (plot.equals("pg3")) {
        set("colorexpr", farField_expr_normalized);
        set("directivityexpr", farField_dir);
    }
    endwith();
}

// -----
// 6. Set Array Dataset and Target Card
// -----

String[] arrPlots = {"arr1", "arr2", "arr3"};
for (String arr : arrPlots) {
    with(model.result().dataset(arr));
    set("fullsize", new String[] {sNarrayx, sNarrayy, "1"});
    set("displ", new String[] {sSpacingx, sSpacingy, "0"});
    endwith();
}

// -----
// 7. Check S-parameter and Plot Far Field
// -----

double sParam = model.result().numerical("gev1").getReal()[0][0];
targetCard = (sParam < dS11Target) ? "passed" : "failed";

```

```

if (!model.sol("sol1").isEmpty() && isArrayViewUpdated && isAntennaUpdated) {
    consoleCard = "allavailable";
} else if (isAntennaUpdated && !isArrayViewUpdated) {
    consoleCard = "clicktoupdate";
} else {
    consoleCard = "notavailable";
}

```

```

model.result("pg3").run();
dDirectivitydB = model.result().table("rp1").getReal()[0][3];
dMaxDirTheta = model.result().table("rp1").getReal()[0][0];
dMaxDirPhi = model.result().table("rp1").getReal()[0][1];

```

```

directivityDisplayText = String.format("Directivity: %.2f (dBi), maximum at  $\varphi$ :  
%.2f (deg),  $\theta$ : %.2f (deg)",  
    dDirectivitydB, dMaxDirPhi, dMaxDirTheta);

```

```

// Reset State Flags

```

```

isGeometryUpdated = false;
isAntennaUpdated = false;
isArrayViewUpdated = false;

```

```

infoCard = "inputchanged";
consoleCard = "notavailable";

```

```

updateArrayResultsButton.set("enabled", false);
reportButton.set("enabled", false);

```

MATLAB CODE FOR PHASE SHIFT AND SNR

% Intelligent Reflecting Surface (IRS) Simulation in MATLAB

clc; clear; close all;

%% Parameters

N = 64; % Number of IRS elements

M = 4; % Number of BS antennas

Pt = 1; % Transmit power (normalized)

noise_power = 0.01; % Noise power at receiver

%% Channel Models

% H_bs_irs: BS to IRS (N x M)

H_bs_irs = (randn(N, M) + 1j * randn(N, M)) / sqrt(2);

% H_irs_ue: IRS to UE (N x 1)

H_irs_ue = (randn(N, 1) + 1j * randn(N, 1)) / sqrt(2);

% H_direct: Direct BS to UE (M x 1)

H_direct = (randn(M, 1) + 1j * randn(M, 1)) / sqrt(2);

%% IRS Phase Optimization (Passive Beamforming)

% Basic phase alignment to maximize signal

theta = angle(H_irs_ue) + angle(H_bs_irs(:,1));

phi = exp(1j * theta); % IRS phase vector

Phi = diag(phi); % IRS matrix

%% Beamforming at BS

% Effective channel through IRS

H_eff_irs = (H_irs_ue') * Phi * H_bs_irs; % 1xM

```

% Matched filter (MRT)
w_irs = H_eff_irs' / norm(H_eff_irs);

% Effective SNR through IRS
SNR_irs = abs(H_eff_irs * w_irs)^2 * Pt / noise_power;
disp(['SNR with IRS: ', num2str(10*log10(SNR_irs)), ' dB']);

%% Direct Transmission (No IRS)
w_direct = H_direct / norm(H_direct);
SNR_direct = abs(H_direct' * w_direct)^2 * Pt / noise_power;
disp(['SNR without IRS: ', num2str(10*log10(SNR_direct)), ' dB']);

%% Plot IRS Phase Shifts
figure;
polarplot(angle(phi), ones(N,1), 'ro');
title('IRS Phase Shifts');
set(gca, 'FontSize', 12);

%% Capacity Comparison (bps/Hz)
capacity_irs = log2(1 + SNR_irs);
capacity_direct = log2(1 + SNR_direct);

fprintf('\nCapacity with IRS: %.2f bps/Hz\n', capacity_irs);
fprintf('Capacity without IRS: %.2f bps/Hz\n', capacity_direct);

%% Optional: Vary IRS size and plot performance
N_list = 4:4:128;
snr_list = zeros(size(N_list));
for idx = 1:length(N_list)
    N_temp = N_list(idx);

```

```

H1 = (randn(N_temp, M) + 1j*randn(N_temp, M))/sqrt(2);
H2 = (randn(N_temp, 1) + 1j*randn(N_temp, 1))/sqrt(2);
theta_temp = angle(H2) + angle(H1(:,1));
phi_temp = exp(1j * theta_temp);
Phi_temp = diag(phi_temp);
H_eff = H2' * Phi_temp * H1;
w = H_eff' / norm(H_eff);
snr_temp = abs(H_eff * w)^2 * Pt / noise_power;
snr_list(idx) = 10*log10(snr_temp);
end

```

```

figure;
plot(N_list, snr_list, '-o');
xlabel('Number of IRS Elements (N)');
ylabel('SNR (dB)');
title('SNR vs. IRS Size');
grid on;

```

% IRS Simulation and Visualization

```

N = 40; % Number of IRS elements
incident_power = abs(randn(1, N)).^2 * 1e-3;
direct_power = 1.8;
reflected_power = 0.2;
total_power = direct_power + reflected_power;
snr_values = [25, 41]; % in dB

```

```

figure('Position', [100, 100, 1000, 400]);

```

% SNR Comparison Plot

```

subplot(2,2,1);

```



```

bar(snr_values, 'FaceColor', [0.2 0.6 1]);
set(gca, 'XTickLabel', {'Without IRS', 'With IRS'});
ylabel('SNR (dB)');
title('SNR Comparison: With vs Without IRS');
grid on;

```

```

% Incident Power at IRS Elements
subplot(2,2,2);
stem(1:N, incident_power, 'b', 'filled');
xlabel('IRS Element Index');
ylabel('Power');
title('Incident Power at Each IRS Element');

```

```

% Power Contributions Plot
subplot(2,2,3);
bar([direct_power, reflected_power, total_power], 'FaceColor', [0.5 0.8 1]);
set(gca, 'XTickLabel', {'Direct', 'Reflected', 'Total'});
ylabel('Power');
title('Power Contributions');

```

```

% Reflected Channel Gain Plot
subplot(2,2,4);
plot(1, 0.95, 'k*', 'MarkerSize', 10);
xlim([0 2]);
ylim([0 1.2]);
xlabel('Sample');
ylabel('|h_{IRS}|');
title('Magnitude of IRS Reflected Channel Gain');
grid on;

```

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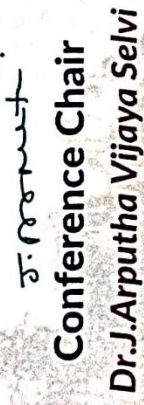
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Convener

Dr.P.P.Shantharaman




Conference Chair

Dr.J.Arputha Vijaya Selvi



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Promotion of Research (CPR), Kings College of Engineering (Autonomous),
Pudukkottai.


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